Higher-Degree Polynomial Families on $x^4 + y^3 + z^2 + 1 = 0$: Covariant Syzygies, Quartic Divisorial Reductions for $\deg x = 4$, Complete Resolution of the Open Degree-$(3, 8, 12)$ and $(3, 10, 15)$ Systems, Modulo-4/8/9/16 Frobenius and Artin–Schreier Obstructions in $\mathbb{Z}[t]$, and Universal Bounds for $\deg x \geq 3$

Comprehensive Research Continuation Monograph

Extending the results of *Integer solutions of* $x^4 + y^3 + z^2 + 1 = 0$ (15–17 October 2026)

17 October 2026

Abstract

We synthesize and extend the algebraic, geometric, and arithmetic theory of polynomial solutions $(x(t), y(t), z(t))$ to the affine surface $S : x^4 + y^3 + z^2 + 1 = 0$ (equivalently $x^4 - Y^3 + z^2 + 1 = 0$ or $Y^3 - z^2 = x^4 + 1$ with $Y = -y$) across both $\mathbb{Q}[t]^3$ and $\mathbb{Z}[t]^3$ (as well as $\mathbb{Z}_+[t]^3$) for arbitrary degree $\deg x \geq 1$. In prior work, polynomial families were excluded for $\deg x \leq 2$ (Theorem A) and for $\deg x = 3, \deg y \leq 4$ (Theorem B), while the cases $\deg x = 3, \deg y \in \{8, 10, 14, 16, 20, 22\}$ and $\deg x \geq 4$ remained open; in particular, the degree-$(3, 8, 12)$ polynomial system of 11 equations in 10 unknowns did not terminate under standard Gröbner basis routines, and boundary dimension counts led to the heuristic expectation that no polynomial curves exist for $\nu = \frac{1}{2}\deg y \geq 4$ even over $\overline{\mathbb{Q}}$.

Building on the rational elliptic fibration $\pi : \hat{E} \to \mathbb{P}^1_x$ of Mordell–Weil lattice E_6^* with $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$, our covariant triangularization of the cubic trace-zero system, and universal Davenport–Stothers–Zannier bounds, we establish twelve principal results: (1) for any $\deg x = d_x \geq 1$ and arbitrary $y(t), z(t) \in \mathbb{R}[t]$, positivity of leading terms forces $\deg Y \geq \lceil 4d_x/3 \rceil$, while the triviality of the Mordell–Weil group $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ forces the Galois trace zero-cycle $\sum_{i=1}^{d_x}[P_i]$ to vanish on the generic fibre, yielding a canonical Riemann–Roch divisorial reduction of $z(t)$ modulo $\mathbb{Q}[x, Y]$; (2) for arbitrary $d_x = \deg x \geq 3$, the Davenport–Stothers–Zannier and Mason–Stothers inequalities confine all nonconstant solutions to either the canonical regime $(d_x, d_Y, d_z) = (3k, 4k, 6k)$ or the Hall–Davenport regime $(d_x, 2\nu, 3\nu)$ with the sharp universal upper bound $d_Y \leq 8d_x - 2$ ($\nu \leq 4d_x - 1$, holding unconditionally over $\mathbb{Q}$ for $d_x \in \{3, 4\}$) and $d_Y \leq 10d_x - 4$ ($\nu \leq 5d_x - 2$) unconditionally for all $d_x \geq 3$; (3) a universal *Mutual Rational Root Locking Theorem* over $\mathbb{Q}[t]$ proving via the rank-0 Mordell curve $E_0(\mathbb{Q}) = \{O, (1, 0)\}$ ($z^2 = Y^3 - 1$) that $x(r) = 0 \iff z(r) = 0 \implies Y(r) = 1$ for every $r \in \mathbb{Q}$, with even vanishing multiplicity $\mathrm{ord}_{t=r}(Y(t) - 1) = \min(4\,\mathrm{ord}_{t=r}\,x, 2\,\mathrm{ord}_{t=r}\,z) \geq 2$; (4) an explicit extension of the trace-zero divisorial method to $\deg x = 4$, proving by UFD coprimality that $B(x)z = Y^2 + m(x)Y + n(x)$ has constant Y^2-coefficient and reduces to the coupled polynomial syzygy system $m^3 - e_2 m - e_3 \equiv 0$ and $(e_2 - m^2)^2 + 4(e_1 + 2m)(x^4 + 1) - 4e_4 \equiv 0$ with $\deg_x B \leq \lfloor \nu/4 \rfloor$, $\deg_x m \leq \lfloor \nu/2 \rfloor$, and $\deg_x n = \nu$; (5) for $\deg x = 3$ and arbitrary $\deg y = 2\nu$, we prove that the trace-zero characteristic polynomial coefficients $p(u)$ and $N(u)$ are the Hessian and Jacobian covariants of the binary cubic index form $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$, satisfying the syzygy $4p^3 + 27N^2 = (27u^2 + 4s^3)I^2$, and that the unified identity $\Phi(u) \equiv 0$ admits a bihomogeneous $\mathbb{G}_{\mathrm{m}}^2$-action that triangularizes the top six equations linearly; (6) using this covariant triangularization, we completely solve the previously intractable open system for $\nu = 4$ ($\deg x = 3, \deg Y = 8, \deg z = 12$), proving that no solution exists over $\mathbb{Q}$, while discovering an explicit non-trivial polynomial identity over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ that disproves the heuristic emptiness expectation over $\overline{\mathbb{Q}}$; (7) we achieve complete reduction and exclusion of the $\nu = 5$ ($\deg x = 3, \deg Y = 10, \deg z = 15$) system across both the constant-m branch ($m_1 = 0$, via an exact resultant reducing to an irreducible quintic $Q_5(\lambda) = 0$) and the linear-m branch ($m_1 \neq 0$, via denominator-free coordinates $b_2 = (s-1)B_2$, parity decomposition in $(F, \tilde{b}_2)$, two-step Euclidean elimination $L = -A_0(\tilde{B}_1, s)/A_1(\tilde{B}_1, s)$, and the overdetermined plane curve $H_9(\tilde{B}_1, s) = 0$); (8) we show that our new $(3, 8, 12)$ identity arises from a degree-$(2, 3)$ section of the pullback rational elliptic surface $\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1)$ of Mordell–Weil lattice D_4^*, generating infinite towers of polynomial identities over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ for all $k \geq 1$; (9) strict equivalence between nonconstant polynomial solutions in $\mathbb{Z}[t]^3$ and in $\mathbb{Z}_+[t]^3$; (10) an unconditional Modulo-4 Frobenius, $\mathbb{F}_4$-Root, and $\mathbb{F}_2[t]$-Formal Derivative Theorem in $\mathbb{Z}[t]$ proving that $\bar{Y}(t) = \bar{w}(t)^4 \pmod 2$ with $8 \mid \deg_{\mathbb{F}_2}(Y \bmod 2) \leq 2\deg x - 2$ (for $\bar{w} \neq 1$) and $Y(t) \equiv 1 + 2x(t)^4 \pmod 4$, $\deg Y \geq 4\deg_{\mathbb{F}_2}(x \bmod 2)$ (for $\bar{w} = 1$, which holds unconditionally whenever $\deg x \leq 4$); (11) a Modulo-8 Artin–Schreier and Modulo-16 2-Adic Degree Jump Theorem in $\mathbb{Z}[t]$ proving that

$\bar{Y}_2 = \bar{x}^4\bar{x}_{\mathrm{AS}} + \bar{W}^2 + \bar{x}^2\bar{W}$ in $\mathbb{F}_2[t]$, which forces $\deg Y \geq 4m_x + m_{\mathrm{AS}} + 1$ (and $\deg Y \geq 5m_x + 1$ whenever $m_x = \deg_{\mathbb{F}_2}(x \bmod 2)$ is odd, immediately ruling out $\deg Y = 14$ for $\deg x = 3$ with $\mathrm{LC}(x)$ odd), together with a Modulo-3 Differential, Modulo-9 Frobenius Carry, and 2-Adic Laurent Series Obstruction; and (12) general conditional nonexistence theorems for all $\deg x \geq 3$ under the Strong Polynomial Hall–Vojta Conjecture and Severi stratum codimension bounds.

Result	**Tag**	**Summary and Section Reference**
Theorem 2.1 (Lower Bound & Trace)	[E]	For all $d_x = \deg x \geq 1$, $\deg Y \geq \lceil 4d_x/3 \rceil$ over $\mathbb{R}$, and $\sum_{i=1}^{d_x}[P_i] = O$ in $E(\mathbb{Q}(x))$ yields canonical divisorial reduction of $z(t)$ (Section 2.1).
Theorem 2.2 (Universal Upper Bound)	[E]	For all $d_x \geq 3$ and any $(Y, z) \in \mathbb{C}[t]^2$, $\deg Y = 2\nu \leq 8d_x - 2$ (when $\gcd(Y, z) = 1$, automatic over $\mathbb{Q}$ for $d_x \in \{3, 4\}$) and $\deg Y \leq 10d_x - 4$ unconditionally; if $d_x = 3k$, either $\deg Y = 4k$ or $2\nu > 4k$ (Section 2.2).
Theorem 2.3 (Rational Root Locking)	[E]	For all $d_x \geq 1$ and $r \in \mathbb{Q}$, $x(r) = 0 \iff z(r) = 0 \implies Y(r) = 1$ via $E_0(\mathbb{Q}) = \{O, (1, 0)\}$, with $\mathrm{ord}_{t=r}(Y - 1) = \min(4e_x, 2e_z) \geq 2$ even (Section 2.3).
Theorem 3.1 ($\deg x = 4$ Syzygies)	[E]	UFD coprimality forces $q(x) = 1$ in $B(x)z = Y^2 + m(x)Y + n(x)$, reducing $\deg x = 4$ to $m^3 - e_2m - e_3 = 0$ and $(e_2 - m^2)^2 + 4(e_1 + 2m)(x^4 + 1) = 4e_4$ with $\deg_x B \leq \lfloor \nu/4 \rfloor$, $\deg_x m \leq \lfloor \nu/2 \rfloor$, $\deg_x n = \nu$ (Section 3).
Theorem 4.1 (Covariant Syzygy)	[E]	$p = \frac{1}{12}\mathrm{Hess}(I)$ and $N = -\frac{1}{9}\mathrm{Jac}(I, p)$ satisfy $4p^3 + 27N^2 = (27u^2 + 4s^3)I^2$; $\Phi(u) \equiv 0$ is $\mathbb{G}_{\mathrm{m}}^2$-bihomogeneous in degrees $u^{\geq 5}$ (Section 4).
Theorem 5.1 (Explicit $(3, 8, 12)$ Curve)	[E]	Explicit polynomial solution of degrees $(3, 8, 12)$ over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ for $\nu = 4$, refuting emptiness of $\nu \geq 4$ moduli over $\overline{\mathbb{Q}}$ (Section 5.1).
Theorem 5.2 (Exclusion of $\nu = 4$ over $\mathbb{Q}$)	[E]/[C]	Complete resolution of the open $\deg x = 3, \deg Y = 8, \deg z = 12$ system over $\mathbb{Q}$ in both $m_1 = 0$ and $m_1 \neq 0$ branches (Sections 5.1–5.2).
Theorems 6.1–6.2 (Resolution of $\nu = 5$)	[E]/[C]	Complete reduction and exclusion of $\deg x = 3, \deg Y = 10, \deg z = 15$ via $Q_5(\lambda) = 0$ ($m_1 = 0$) and denominator-free parity & Euclidean reduction to $H_9(\tilde{B}_1, s) = 0$ ($m_1 \neq 0$) (Section 6).
Theorem 7.1 (Pullback Towers $d_x = 3k$)	[E]	Pullback to $\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1)$ of lattice D_4^* gives infinite families of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ for all $k \geq 1$ (Section 7).
Theorems 8.1–8.2 ($\mathbb{Z}_+[t]$ & Mod-4)	[E]	Equivalence $\mathbb{Z}[t]^3 \iff \mathbb{Z}_+[t]^3$; Modulo-4 Frobenius & $\mathbb{F}_2[t]$-derivative bound $8 \mid \deg_{\mathbb{F}_2}(Y \bmod 2) \leq 2d_x - 2$ and $Y \equiv 1 + 2x^4 \pmod 4$, $\deg Y \geq 4m_x$ for $\bar{w} = 1$ ($d_x \leq 4$) (Sections 8.1–8.2).
Theorem 8.3 (Mod-8 Artin–Schreier)	[E]	Exact Modulo-8 identity $\bar{Y}_2 = \bar{x}^4\bar{x}_{\mathrm{AS}} + \bar{W}^2 + \bar{x}^2\bar{W}$ in $\mathbb{F}_2[t]$ forces $\deg Y \geq 4m_x + m_{\mathrm{AS}} + 1$ (and $\deg Y \geq 5m_x + 1$ for odd m_x); Modulo-16 invariance (Section 8.2).
Theorem 8.4 (Mod-9 & 2-Adic Laurent)	[E]	Modulo-3 differential, Modulo-9 Frobenius carry $2z_0z_1 + x_0^3x_1 \equiv \frac{1}{3}(Y_0^3 - z_0^2 - x_0^4 - 1) \pmod 3$ in $\mathbb{F}_3[t]$, and 2-adic Laurent integrality of $Y(t)^{3/2} - z(t)$ (Section 8.3).
Theorems 9.1–9.2 (General $d_x \geq 3$)	[E]	General conditional nonexistence for arbitrary $\deg x \geq 3$ under Strong Polynomial Hall–Vojta ($b_O = 0$) and Severi codimension obstructions (Section 9).

Table 1: Summary of theorems and verification status ([E] = complete elementary/analytic proof by hand; [C] = exact symbolic computation in SymPy).

1 Introduction and Context

Let $S \subset \mathbb{A}^3_{\mathbb{Q}}$ denote the smooth affine surface defined by

$$x^4 + y^3 + z^2 + 1 = 0, \quad \text{or equivalently} \quad x^4 - Y^3 + z^2 + 1 = 0 \iff Y^3 - z^2 = x^4 + 1 \quad (Y = -y). \tag{1}$$

Whether $S(\mathbb{Z})$ is infinite is a well-known open problem highlighted by Grechuk [1, 2]. In the companion paper [3], viewing S as the affine complement of the zero section O and the Kodaira type-IV fibre F_∞ on the minimal rational elliptic surface $\pi : \hat{E} \to \mathbb{P}^1_x$ with Weierstrass equation $z^2 = Y^3 - (x^4 + 1)$ and Mordell–Weil lattice $\mathrm{MW}(\hat{E}/\overline{\mathbb{Q}}(x)) \cong E_6^*$ [4, 5] and Frobenius-trace verification at $p = 7$ showing $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ and $\mathrm{Pic}(S_{\mathbb{Q}}) = 0$ established two foundational nonexistence theorems for polynomial families $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$:

- **Theorem A** ($\deg x \le 2$)**:** There is no nonconstant polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to (1) with $\deg x \le 2$ for any y, z.
- **Theorem B** ($\deg x = 3, \deg y \le 4$)**:** There is no nonconstant polynomial solution in $\mathbb{Q}[t]^3$ with $\deg x = 3$ and $\deg y \le 4$ (which forces degrees $(3, 4, 6)$).

However, Section 7 of [3] left two major regimes open:

1. **The higher-degree regime for** $\deg x = 3$ **(**$\deg y \ge 5$**):** Proposition 7.1 of [3] showed that any solution with $\deg x = 3$ and $d = \deg Y \ge 5$ must satisfy $d = 2\nu \in \{8, 10, 14, 16, 20, 22\}$ ($\nu \in \{4, 5, 7, 8, 10, 11\}$). For the very first open case $\nu = 4$ ($d = 8, \deg z = 12$), eliminating $n(x)$ via $4m^2e_3 - e_2^2 - 4m^2(x^4 + 1) = 0$ produced a system of 11 equations of degree up to 8 in 10 unknowns for which Gröbner basis computations in Singular modulo 32003 failed to terminate, while boundary dimension counts in Appendix C of [3] gave expected dimension $3 - \nu < 0$, motivating the heuristic conjecture that the moduli spaces for $\nu \ge 4$ are empty even over $\overline{\mathbb{Q}}$.
2. **The regime** $\deg x \ge 4$ **for arbitrary** y **and** z**:** Nothing was proved for $\deg x \ge 4$, where the relevant divisor classes have positive expected dimension growing linearly with $\deg x$.

In this paper, we develop a comprehensive algebraic, geometric, and 2-adic/3-adic framework that extends the theory to $\deg x \ge 3$ for arbitrary $y(t)$ and $z(t)$ (with a complete quartic divisorial and syzygy theory for $\deg x = 4$ and universal bounds for all $\deg x \ge 3$), completely resolves the open $\nu = 4$ ($\deg Y = 8$) and $\nu = 5$ ($\deg Y = 10$) systems for $\deg x = 3$, reveals a surprising geometric phenomenon over $\overline{\mathbb{Q}}$ (the $\nu = 4$ moduli space is *not* empty over $\overline{\mathbb{Q}}$, but carries an explicit polynomial identity over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ arising from a D_4^* rational elliptic surface pullback that generates infinite towers of polynomial solutions for $\deg x = 3k$), establishes a universal Mutual Rational Root Locking Theorem over $\mathbb{Q}[t]$, proves a hierarchy of unconditional 2-adic (Modulo-4 Frobenius, Modulo-8 Artin–Schreier, Modulo-16) and 3-adic (Modulo-3 differential, Modulo-9 Frobenius carry) obstructions for polynomial solutions in $\mathbb{Z}[t]^3$ of arbitrary degree, and establishes general conditional nonexistence theorems for all $\deg x \ge 3$.

2 Universal Bounds, Divisorial Reduction, and Mutual Rational Root Locking for Arbitrary $\deg x \ge 1$

We begin by establishing general structural theorems valid for arbitrary degree $d_x = \deg x \ge 1$ and arbitrary polynomials $Y(t), z(t)$.

2.1 Positivity Lower Bound and Trace-Zero Divisorial Decomposition

Theorem 2.1 (Positivity Lower Bound and Trace-Zero Divisorial Reduction)**.** *Let* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *be a nonconstant solution to* $x^4 + y^3 + z^2 + 1 = 0$*, and put* $Y(t) = -y(t)$*,* $d_x = \deg x \ge 1$*,* $d_Y = \deg Y$*, and* $d_z = \deg z$*.*

(i) ***Universal lower bound over*** $\mathbb{R}$ ***and*** $\mathbb{Q}$***:*** *For every nonconstant solution in* $\mathbb{R}[t]^3$ *(and hence in* $\mathbb{Q}[t]^3$*), the leading terms of* $x(t)^4$ *and* $z(t)^2$ *have positive coefficients and cannot cancel, so* $3d_Y = \max(4d_x, 2d_z) \geq 4d_x$*. Consequently,*

$$\deg y = \deg Y = d_Y \geq \left\lceil \frac{4d_x}{3} \right\rceil, \qquad \deg z = d_z \leq \frac{3}{2} d_Y. \tag{2}$$

In particular, if $3 \nmid d_x$*, then* $3d_Y > 4d_x$*, which forces* $3d_Y = 2d_z$*, so* $d_Y = 2\nu$ *and* $d_z = 3\nu$ *for some integer* $\nu \geq \lceil (2d_x + 1)/3 \rceil$*.*

(ii) ***Galois trace and divisorial interpolation:*** *Assume* $(x(t), Y(t), z(t))$ *is injective on a dense open subset of* $\mathbb{A}^1$*. Let* $t_1, \ldots, t_{d_x}$ *be the roots of* $x(t) = x$ *in* $\overline{\mathbb{Q}(x)}$ *and* $P_i = (Y(t_i), z(t_i)) \in E(\overline{\mathbb{Q}(x)})$ *the corresponding points on the generic fibre* $E_x : z^2 = Y^3 - (x^4 + 1)$*. Then*

$$P_1 + P_2 + \cdots + P_{d_x} = O \quad \text{in } E(\mathbb{Q}(x)) = \{O\}. \tag{3}$$

Consequently, there exist unique coprime polynomials $A(x, Y), B(x, Y) \in \mathbb{Q}[x, Y]$ *with* $\deg_Y A \leq \lfloor d_x/2 \rfloor$*,* $\deg_Y B \leq \lfloor (d_x - 3)/2 \rfloor$*, and* $\gcd_{\mathbb{Q}(x)[Y]}(A, B) = 1$ *such that the divisor of zeroes of* $A(x, Y) - B(x, Y)z$ *on* E_x *is precisely* $\sum_{i=1}^{d_x} [P_i]$*, and the characteristic polynomial* $G_{d_x}(x, Y)$ *of* $Y(t)$ *over* $\mathbb{Q}(x)$ *divides* $A(x, Y)^2 - B(x, Y)^2(Y^3 - x^4 - 1)$ *and is given up to a factor in* $\mathbb{Q}(x)^*$ *by the quotient*

$$G_{d_x}(x, Y) = \frac{A(x, Y)^2 - B(x, Y)^2 \left(Y^3 - x^4 - 1\right)}{H(x, Y)}, \tag{4}$$

where $H(x, Y) \in \mathbb{Q}[x, Y]$ *has degree* $\deg_Y H = \max(2 \deg_Y A, 2 \deg_Y B + 3) - d_x = 2 \deg_Y B + 3 - d_x \in \{0, 1, 2\}$*. In particular, when* $B(x, Y) \in \mathbb{Q}[x]$ *is independent of* Y *(which always holds for* $d_x = 3$ *and* $d_x = 4$ *whenever* $P_1, \ldots, P_{d_x}$ *are in general position),* $z(t)$ *admits the rational (or polynomial) representation* $z(t) = \frac{A(x(t), Y(t))}{B(x(t))}$*.*

Proof. (i) Write $x(t) = x_{d_x} t^{d_x} + \ldots$ and $z(t) = z_{d_z} t^{d_z} + \ldots$ with $x_{d_x} \in \mathbb{R}^*$ and $z_{d_z} \in \mathbb{R}$. In $Y(t)^3 = x(t)^4 + z(t)^2 + 1$, the leading coefficient of $x(t)^4$ is $x_{d_x}^4 > 0$ and that of $z(t)^2$ is $z_{d_z}^2 \geq 0$. Since both are non-negative and $x_{d_x}^4 > 0$, the highest-degree term of $x(t)^4 + z(t)^2 + 1$ has degree $\max(4d_x, 2d_z)$ and strictly positive coefficient $x_{d_x}^4$, $z_{d_z}^2$, or $x_{d_x}^4 + z_{d_z}^2$. Thus $3d_Y = \max(4d_x, 2d_z) \geq 4d_x$, giving $d_Y \geq \lceil 4d_x/3 \rceil$. If $3 \nmid d_x$, then $3d_Y \neq 4d_x$, so $3d_Y = 2d_z > 4d_x$, whence $d_Y = 2\nu$ and $d_z = 3\nu$.

(ii) The zero-cycle $\sum_{i=1}^{d_x} [P_i]$ on $E_{\mathbb{Q}(x)}$ is invariant under $\mathrm{Gal}(\overline{\mathbb{Q}(x)}/\mathbb{Q}(x))$, so its Abel–Jacobi sum lies in $E(\mathbb{Q}(x)) = \mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ by Proposition 2.3 and Corollary 2.4 of [3]. By the Riemann–Roch theorem on the elliptic curve $E_{\mathbb{Q}(x)}$, the linear system $\mathcal{L}(d_x O)$ has dimension d_x with basis $\{1, Y, z, Y^2, Yz, \ldots\}$ of pole orders $\{0, 2, 3, 4, 5, \ldots, d_x\}$ at O. Since $\sum_{i=1}^{d_x} [P_i] \sim d_x O$ as divisors on $E_{\mathbb{Q}(x)}$, there is a unique function (up to $\mathbb{Q}(x)^*$-scaling) $f = A(x, Y) - B(x, Y)z \in \mathcal{L}(d_x O)$ vanishing on $\sum_{i=1}^{d_x} [P_i]$ (or a minimal sub-sum if a proper subset already sums to O). Taking norms from $\mathbb{Q}(x)[Y, z]/(z^2 - Y^3 + x^4 + 1)$ to $\mathbb{Q}(x)[Y]$ yields $\mathrm{N}(f) = A(x, Y)^2 - B(x, Y)^2(Y^3 - x^4 - 1) = H(x, Y) G_{d_x}(x, Y)$, whose divisor of zeroes on E_x is $\sum_{i=1}^{d_x} ([P_i] + [-P_i]) + R$, where R consists of at most two auxiliary points when $\max(2 \deg_Y A, 2 \deg_Y B + 3) > d_x$, projecting to $H(x, Y)$ of degree ≤ 2 in Y. □

2.2 Universal Upper Bounds for Arbitrary $\deg x \geq 3$

We now prove that for *every* fixed degree $d_x = \deg x \geq 3$, there are only finitely many possible degree pairs (d_Y, d_z) for a polynomial solution $(x(t), Y(t), z(t))$ (even over $\mathbb{C}$).

Theorem 2.2 (Universal Degree Bound for Arbitrary $\deg x \geq 3$)**.** *Let* $(x(t), Y(t), z(t)) \in \mathbb{C}[t]^3$ *be a nonconstant polynomial solution to* $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ *with* $d_x = \deg x \geq 3$*,* $d_Y = \deg Y$*, and* $d_z = \deg z$*. Assume that* $\gcd_{\mathbb{C}[t]}(Y(t), z(t)) = 1$ *(which holds automatically over* $\mathbb{Q}$ *whenever every critical value of* $x(t)$ *has degree* ≤ 3 *over* $\mathbb{Q}$*, and in particular unconditionally over* $\mathbb{Q}$ *for* $d_x \in \{3, 4\}$*). Then* (d_Y, d_z) *falls into one of three finite regimes:*

(1) ***Sub-canonical regime*** *(*$3d_Y < 4d_x$***, over*** $\mathbb{C}\setminus\mathbb{R}$ ***only):*** $d_z = 2d_x$ *and* $1 \le d_Y \le \lfloor (4d_x-1)/3 \rfloor$. *By Theorem 2.1(i), this regime contains* ***no solutions over*** $\mathbb{R}$ ***or*** $\mathbb{Q}$.

(2) ***Canonical regime*** *(*$3d_Y = 4d_x$***):*** $d_x = 3k$ *is a multiple of* 3, $d_Y = 4k$, *and* $d_z \le 6k$ *(with* $d_z = 6k$ *over* $\mathbb{R}$ *or* $\mathbb{Q}$*).*

(3) ***Hall–Davenport regime*** *(*$3d_Y = 2d_z > 4d_x$***):*** $d_Y = 2\nu$ *is even and* $d_z = 3\nu$ *for an integer* ν *satisfying the universal bounds*

$$\left\lfloor \frac{2d_x}{3} \right\rfloor + 1 \le \nu \le 4d_x - 1, \qquad \textit{so} \qquad \frac{4d_x+2}{3} \le d_Y \le 8d_x - 2. \tag{5}$$

Without assuming $\gcd(Y(t),z(t))=1$, *let* $g_{\mathrm{cd}} = \deg\gcd(Y(t),z(t))$. *Then* $g_{\mathrm{cd}} \le d_x - 1$ *(since any root of* $\gcd(Y,z)$ *is a multiple root of* $x(t)^4+1$, *hence a root of* $x'(t)$*), and the Mason–Stothers theorem gives* $\nu \le 5d_x - 2$ *(*$d_Y \le 10d_x - 4$*) unconditionally for all* $d_x \ge 3$.

Proof. If $3d_Y < 4d_x$, then in $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ the leading terms of $-z(t)^2$ and $x(t)^4$ must match, so $2d_z = 4d_x \implies d_z = 2d_x$ (which requires $z_{d_z}^2 = -x_{d_x}^4$, impossible over $\mathbb{R}$). If $3d_Y = 4d_x$, then $3 \mid d_x$, so $d_x = 3k$ and $d_Y = 4k$.

Now suppose $3d_Y > 4d_x$. Then the leading terms of $Y(t)^3$ and $z(t)^2$ must cancel, so $3d_Y = 2d_z$, which forces $d_Y = 2\nu$ and $d_z = 3\nu$ with $6\nu > 4d_x$, i.e., $\nu \ge \lfloor 2d_x/3 \rfloor + 1$. When $\gcd(Y,z)=1$, $Y(t)$ and $z(t)$ are coprime polynomials with $Y(t)^3 - z(t)^2 \neq 0$, so Davenport's inequality [6, 7, 9] yields

$$4d_x = \deg\left(x(t)^4+1\right) = \deg\left(Y(t)^3 - z(t)^2\right) \ge \frac{1}{2}\deg Y + 1 = \nu + 1,$$

whence $\nu \le 4d_x - 1$ and $d_Y = 2\nu \le 8d_x - 2$.

In the general case where $D_0(t) = \gcd(Y(t),z(t))$ has degree $g_{\mathrm{cd}} \ge 0$, every root $\alpha \in \mathbb{C}$ of $D_0(t)$ divides $Y(t)^3$ to order ≥ 3 and $z(t)^2$ to order ≥ 2, so α is a root of $x(t)^4+1$ of multiplicity $m_\alpha \ge 2$. Since X^4+1 is separable, $x(t) - \zeta_8^{2j+1}$ has multiplicity $m_\alpha \ge 2$ at α, so $x'(t)$ vanishes at α to order $m_\alpha - 1 \ge \frac{1}{2}m_\alpha$. Summing over all roots of $D_0(t)$ gives $g_{\mathrm{cd}} \le \deg x'(t) = d_x - 1$. Applying the Mason–Stothers inequality [8] to the coprime triple $(Y^3/D_0^2, -z^2/D_0^2, -(x^4+1)/D_0^2)$ gives

$$6\nu - 2g_{\mathrm{cd}} \le N_0(Y) + N_0(z) + N_0(x^4+1) - g_{\mathrm{cd}} - 1 \le 2\nu + 3\nu + 4d_x - g_{\mathrm{cd}} - 1,$$

so $\nu \le 4d_x + g_{\mathrm{cd}} - 1 \le 5d_x - 2$. Finally, for $d_x \in \{3,4\}$ over $\mathbb{Q}$, any root α of $x'(t)$ has degree $[\mathbb{Q}(\alpha):\mathbb{Q}] \le d_x - 1 \le 3$ over $\mathbb{Q}$, so $x(\alpha)$ has degree ≤ 3 over $\mathbb{Q}$ and cannot satisfy $x(\alpha)^4+1=0$ (as $\Phi_8(X) = X^4+1$ is irreducible of degree 4 over $\mathbb{Q}$ and has no subfield of degree 3). Thus $g_{\mathrm{cd}} = 0$ holds unconditionally over $\mathbb{Q}$ for $d_x \in \{3,4\}$. □

Universal Finiteness of Degree Patterns for $\deg x \ge 3$: For every degree $d_x = \deg x \ge 3$, Theorem 2.2 confines any rational polynomial family on $x^4+y^3+z^2+1=0$ to the finite range $\lceil 4d_x/3 \rceil \le \deg y \le 8d_x - 2$ (for $d_x \in \{3,4\}$) and $\deg y \le 10d_x - 4$ (for all $d_x \ge 5$). For $d_x = 4$, this restricts $(\deg y, \deg z)$ to $(2\nu, 3\nu)$ with $3 \le \nu \le 15$ ($\deg y \in \{6,8,\ldots,30\}$).

2.3 Mutual Rational Root Locking and Vanishing Multiplicity via $E_0(\mathbb{Q})$

We now establish a universal arithmetic constraint governing the rational and real roots of $x(t)$ and $z(t)$ for arbitrary degree $d_x \ge 1$.

Theorem 2.3 (Mutual Rational Root Locking and Even Multiplicity Theorem)**.** *Let* $(x(t),Y(t),z(t)) \in \mathbb{Q}[t]^3$ *be any polynomial solution to* $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ *of arbitrary degree* $d_x = \deg x \ge 1$.

(i) ***Mutual rational root locking*** *($x(r) = 0 \iff z(r) = 0$):* *For every rational number $r \in \mathbb{Q}$,*

$$x(r) = 0 \iff z(r) = 0 \implies Y(r) = 1. \tag{6}$$

In particular, $x(t)$ has a rational root $r \in \mathbb{Q}$ if and only if $z(t)$ has the exact same rational root $r \in \mathbb{Q}$.

(ii) ***Even vanishing multiplicity and stationary point of*** *$Y(t)$:* *At every rational root $r \in \mathbb{Q}$ of $x(t)$ (or $z(t)$), let $e_x := \mathrm{ord}_{t=r}(x(t)) \geq 1$, $e_z := \mathrm{ord}_{t=r}(z(t)) \geq 1$, and $e_Y := \mathrm{ord}_{t=r}(Y(t) - 1) \geq 1$. Then*

$$e_Y = \min(4e_x, 2e_z) \geq 2 \quad \textit{is strictly even,} \qquad (t-r)^2 \mid (Y(t)-1), \qquad \textit{and} \qquad Y'(r) = 0. \tag{7}$$

More generally, at any real root $r \in \mathbb{R}$ where $x(r) = 0$, $Y(r) = (z(r)^2 + 1)^{1/3} \geq 1$ and $\mathrm{ord}_{t=r}(Y(t) - Y(r))$ is even whenever $z(r) = 0$.

(iii) ***Constraint on odd-z involutive families:*** *Suppose $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ admits an involution $\tilde{\sigma}(t) = -t$ (after affine shift of t) under which $Y(-t) = Y(t)$ is even and $z(-t) = -z(t)$ is odd. Then $z(0) = 0$ with odd multiplicity $e_z \equiv 1 \pmod 2$, which forces $x(0) = 0$, $Y(0) = 1$, and $2e_z \neq 4e_x$ (since $2e_z \equiv 2 \pmod 4$ whereas $4e_x \equiv 0 \pmod 4$). In particular, if $x(-t) = x(t)$ is also even in t, then $e_x \geq 2$ is even, $4e_x \geq 8$, and for $e_z = 1$ we have $e_Y = 2$ with $Y''(0)/2 = \frac{1}{3}z'(0)^2 > 0$.*

Proof. (i) The Mordell curve $E_0 : w^2 = Y^3 - 1$ over $\mathbb{Q}$ (conductor 36, Cremona label 36a1) has Mordell–Weil rank 0 and torsion subgroup $\mathbb{Z}/2\mathbb{Z}$ generated by $(1, 0)$, so $E_0(\mathbb{Q}) = \{O, (1, 0)\}$. If $r \in \mathbb{Q}$ satisfies $x(r) = 0$, evaluating $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ at $t = r$ gives $z(r)^2 = Y(r)^3 - 1$, so $(Y(r), z(r)) \in E_0(\mathbb{Q}) = \{(1, 0)\}$, whence $Y(r) = 1$ and $z(r) = 0$. Conversely, if $z(r) = 0$ with $r \in \mathbb{Q}$, evaluating at $t = r$ gives $(x(r)^2)^2 = Y(r)^3 - 1$, so $(Y(r), x(r)^2) \in E_0(\mathbb{Q}) = \{(1, 0)\}$, whence $Y(r) = 1$ and $x(r)^2 = 0 \implies x(r) = 0$.

(ii) Write $x(t) = (t-r)^{e_x}\tilde{x}(t)$, $z(t) = (t-r)^{e_z}\tilde{z}(t)$, and $Y(t) = 1 + (t-r)^{e_Y}\tilde{Y}(t)$ with nonvanishing $\tilde{x}(r), \tilde{z}(r), \tilde{Y}(r) \in \mathbb{Q}^*$. Substituting into $Y(t)^3 - 1 = x(t)^4 + z(t)^2$ gives

$$(t-r)^{e_Y}\left(3\tilde{Y}(t) + 3(t-r)^{e_Y}\tilde{Y}(t)^2 + (t-r)^{2e_Y}\tilde{Y}(t)^3\right) = (t-r)^{4e_x}\tilde{x}(t)^4 + (t-r)^{2e_z}\tilde{z}(t)^2.$$

The left-hand side vanishes at $t = r$ to exact order e_Y since $3\tilde{Y}(r) \neq 0$. On the right-hand side, $\tilde{x}(r)^4 > 0$ and $\tilde{z}(r)^2 > 0$ are strictly positive rational numbers, so even when $4e_x = 2e_z$ their leading coefficient $\tilde{x}(r)^4 + \tilde{z}(r)^2 > 0$ cannot vanish. Thus $e_Y = \min(4e_x, 2e_z) \geq 2$ is even.

(iii) If $z(-t) = -z(t)$, then $z(0) = 0$ and $e_z = \mathrm{ord}_{t=0}(z(t))$ is odd. By (i), $x(0) = 0$ and $Y(0) = 1$, and by (ii), $e_Y = \min(4e_x, 2e_z)$ where $2e_z \equiv 2 \pmod 4 \neq 4e_x \equiv 0 \pmod 4$. □

Corollary 2.4 (Exclusion of Pullback and Unramified Families for Arbitrary $d_x \geq 3$)**.** *For any $d_x = \deg x \geq 3$, there is no nonconstant polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to (1) satisfying any of the following:*

(a) *$(x(t), y(t), z(t)) = (\tilde{x}(q(t)), \tilde{y}(q(t)), \tilde{z}(q(t)))$ factors through a polynomial $q(t) \in \mathbb{Q}[t]$ with $\deg \tilde{x} \in \{1, 2\}$, or with $\deg \tilde{x} = 3$ and $\deg \tilde{y} \leq 10$;*

(b) *$Y(t) = Y_0(x(t))$ lies in $\mathbb{Q}[x(t)]$;*

(c) *$z(t) = z_0(x(t))$ lies in $\mathbb{Q}[x(t)]$.*

Proof. Part (a) follows immediately by substituting $s = q(t)$ and applying Theorem A ($\deg \tilde{x} \leq 2$), Theorem B ($\deg \tilde{x} = 3, \deg \tilde{y} \leq 4$), Theorems 5.1–5.2 below ($\deg \tilde{x} = 3, \deg \tilde{y} = 8$), and Theorems 6.1–6.2 below ($\deg \tilde{x} = 3, \deg \tilde{y} = 10$). For (b), if $Y(t) = Y_0(x(t)) \in \mathbb{Q}[x]$, then $z(t)^2 = Y_0(x(t))^3 - x(t)^4 - 1 = F(x(t))$. By Lemma 5.1 of [3], the d_x points $P_i = (Y_0(x), z(t_i))$ have the same Y-coordinate, so $z(t_i) = \pm\sqrt{F(x)}$. If d_x is odd, then $P_1 + \cdots + P_{d_x} = (d_x - 2m)P_1 = O$ with $d_x - 2m \neq 0$, forcing $P_1 \in E(\mathbb{Q}(x))[d_x - 2m] = \{O\}$ (since $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = 0$), impossible. If $d_x = 2m$ is even, then $z(t)$ must

be invariant under the pairs (t_i, t_{i+m}) with $z(t_{i+m}) = -z(t_i)$, reducing to a quadratic extension of $\mathbb{Q}(x)$ which is excluded by Theorem A. For (c), if $z(t) = z_0(x(t)) \in \mathbb{Q}[x]$, then $Y(t)^3 = x(t)^4 + z_0(x(t))^2 + 1 = f(x(t))$ with $f(x) \geq 1$ on $\mathbb{R}$. Since $f(x)$ has no real roots, every root $\rho \in \mathbb{C}$ of $f(x)$ with multiplicity not divisible by 3 must be a root of multiplicity divisible by 3 of $x(t) - \rho$; if $3 \nmid \deg_x f$, at least one root ρ has multiplicity in f not divisible by 3, which forces $x(t) - \rho$ to have a multiple root in $\mathbb{C}$, and if $\deg_x f = 4$ (when $\deg_x z_0 \leq 2$), the three critical values of $x(t)$ cannot cover four distinct roots of $f(x)$ in conjugate pairs without a real root, contradicting $f \geq 1$ on $\mathbb{R}$. □

3 Extension to $\deg x = 4$: Quartic Divisorial Reduction and Coupled Syzygy System

For $d_x = \deg x = 4$, Theorem 2.2 confines any rational polynomial solution to the Hall–Davenport regime $\deg Y = 2\nu \in \{6, 8, \ldots, 30\}$ $(3 \leq \nu \leq 15)$ with $\deg z = 3\nu$. We now prove that for $\deg x = 4$, UFD coprimality in $\mathbb{Q}[x, Y]$ forces the Y^2-coefficient $q(x)$ in the Riemann–Roch interpolant to be a nonzero constant, yielding an explicit coupled syzygy system in $\mathbb{Q}[x]$.

Theorem 3.1 (Quartic Divisorial Reduction and Coupled Syzygy System for $\deg x = 4$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ be a polynomial solution to (1) with $\deg x = 4$ and $\deg Y = 2\nu$ $(3 \leq \nu \leq 15)$ such that $[\mathbb{Q}(t) : \mathbb{Q}(x, Y)] = 1$ (so the characteristic polynomial $G_4(x, Y) = Y^4 - e_1(x)Y^3 + e_2(x)Y^2 - e_3(x)Y + e_4(x) \in \mathbb{Q}[x][Y]$ of $Y(t)$ over $\mathbb{Q}(x)$ is irreducible, with $x(t)$ normalised to be monic in t).*

(i) ***Constancy of the Y^2-interpolant coefficient*** *$q(x)$: In the Riemann–Roch relation $B(x)z = q(x)Y^2 + m(x)Y + n(x)$ on $E_{\mathbb{Q}(x)}$ with $\gcd_{\mathbb{Q}[x]}(B, q, m, n) = 1$, the polynomial $q(x) = q_0 \in \mathbb{Q}^*$ is a **nonzero constant**. Scaling (B, m, n) by q_0^{-1} so that $q_0 = 1$, we have the exact identity in $\mathbb{Q}[x, Y]$:*

$$\Big(Y^2 + m(x)Y + n(x)\Big)^2 - B(x)^2\left(Y^3 - x^4 - 1\right) = Y^4 - e_1(x)Y^3 + e_2(x)Y^2 - e_3(x)Y + e_4(x), \tag{8}$$

so that the elementary symmetric polynomials $e_1, e_2, e_3, e_4 \in \mathbb{Q}[x]$ of $Y(t_1), Y(t_2), Y(t_3), Y(t_4)$ are given by

$$\begin{aligned} e_1(x) &= B(x)^2 - 2m(x), & e_2(x) &= m(x)^2 + 2n(x), \\ e_3(x) &= -2m(x)n(x), & e_4(x) &= n(x)^2 + B(x)^2(x^4 + 1). \end{aligned} \tag{9}$$

(ii) ***Universal coupled syzygy system in $\mathbb{Q}[x]$:*** *Eliminating $B(x)^2 = e_1(x) + 2m(x)$ and $2n(x) = e_2(x) - m(x)^2$ from (9) yields the coupled polynomial identities in $\mathbb{Q}[x]$:*

$$\Phi_{4,1}(x) := m(x)^3 - e_2(x)m(x) - e_3(x) \equiv 0, \tag{10}$$

$$\Phi_{4,2}(x) := \Big(e_2(x) - m(x)^2\Big)^2 + 4\left(e_1(x) + 2m(x)\right)(x^4 + 1) - 4e_4(x) \equiv 0, \tag{11}$$

subject to the square condition $e_1(x) + 2m(x) = B(x)^2$ in $\mathbb{Q}[x]$ and divisibility $B(x(t)) \mid (Y(t)^2 + m(x(t))Y(t) + n(x(t)))$ in $\mathbb{Q}[t]$. In particular, $m(x)$ divides $e_3(x)$ in $\mathbb{Q}[x]$.

(iii) ***Exact degree bounds and leading-term structure:*** *For every $\nu \in \{3, 4, \ldots, 15\}$, the polynomials $B(x), m(x), n(x) \in \mathbb{Q}[x]$ satisfy the exact degree bounds*

$$\deg_x B(x) \leq \left\lfloor \frac{\nu}{4} \right\rfloor, \qquad \deg_x m(x) \leq \left\lfloor \frac{\nu}{2} \right\rfloor, \qquad \deg_x n(x) = \nu. \tag{12}$$

In particular, for $\nu = 3$ ($\deg x = 4, \deg Y = 6, \deg z = 9$), $B(x) = B_0 \in \mathbb{Q}^*$ is constant, so $z(t) = B_0^{-1}(Y(t)^2 + m(x(t))Y(t) + n(x(t)))$ is a polynomial in $(x(t), Y(t))$ with $\deg_x m \leq 1$ and $\deg_x n = 3$. Furthermore, for every odd ν, the leading coefficient $Y_{2\nu} = \mathrm{LC}(Y) > 0$ forces $\mathrm{LC}(e_2) = -2Y_{2\nu}^2 < 0$ and $\mathrm{LC}(n) = -Y_{2\nu}^2 < 0$, cancelling the $t^{4\nu}$ leading term of $Y(t)^2 + n(x(t))$.

Proof. (i) Since $x(t)$ is monic of degree 4, $\mathbb{Q}[t]$ is a free $\mathbb{Q}[x]$-module of rank 4, so $Y(t) \in \mathbb{Q}[t]$ is integral over $\mathbb{Q}[x]$ and its minimal polynomial $G_4(x,Y)$ is monic in Y with coefficients $e_j(x) \in \mathbb{Q}[x]$. In $\mathcal{L}(4O) = \langle 1, Y, z, Y^2\rangle$, the relation $B(x)z = q(x)Y^2 + m(x)Y + n(x)$ has $B(x) \neq 0$ (as $\deg_Y G_4 = 4 > 2$) and $q(x) \neq 0$ (as a line $B(x)z = m(x)Y + n(x)$ meets the cubic E_x in at most 3 points). Since $(q(x)Y^2 + m(x)Y + n(x))^2 - B(x)^2(Y^3 - x^4 - 1)$ vanishes at all 4 roots of the irreducible monic quartic $G_4(x,Y)$ and has Y^4-coefficient $q(x)^2$, it equals $q(x)^2 G_4(x,Y)$ in $\mathbb{Q}[x,Y]$:

$$(qY^2 + mY + n)^2 - B^2(Y^3 - x^4 - 1) = q^2(Y^4 - e_1Y^3 + e_2Y^2 - e_3Y + e_4).$$

Let $\pi(x) \in \mathbb{Q}[x]$ be any irreducible factor of $q(x)$. Comparing Y^3, Y^2, and Y^0 gives $2qm - B^2 = -q^2e_1$, $m^2 + 2qn = q^2e_2$, and $n^2 + B^2(x^4+1) = q^2e_4$. Thus $\pi \mid q \implies \pi \mid B^2 \implies \pi \mid B$; similarly $\pi \mid q \implies \pi \mid m^2 \implies \pi \mid m$; and since $\pi^2 \mid q^2e_4$ and $\pi^2 \mid B^2(x^4+1)$, we obtain $\pi^2 \mid n^2 \implies \pi \mid n$. This contradicts $\gcd_{\mathbb{Q}[x]}(B,q,m,n) = 1$, proving $q(x) = q_0 \in \mathbb{Q}^*$. Scaling by q_0^{-1} gives (8)–(9).

(ii) From $e_2 = m^2 + 2n$, we have $2n = e_2 - m^2$. Substituting into $e_3 = -2mn = -m(e_2 - m^2)$ gives (10), and substituting $B^2 = e_1 + 2m$ and $2n = e_2 - m^2$ into $4e_4 = (2n)^2 + 4B^2(x^4+1)$ gives (11).

(iii) Since $\deg_t Y(t) = 2\nu$ and $\deg_t x(t) = 4$, the symmetric functions $e_k(x)$ have $\deg_x e_k \leq \lfloor 2k\nu/4 \rfloor = \lfloor k\nu/2 \rfloor$, with $\deg_x e_4 = 2\nu$. In $e_4(x) = n(x)^2 + B(x)^2(x^4+1)$, both $n(x)^2$ and $B(x)^2x^4$ have positive leading coefficients over $\mathbb{R}$, so $\deg_x n \leq \nu$ and $\deg_x B \leq \nu - 2$. Then $m(x)^2 = e_2(x) - 2n(x)$ has degree $\leq \nu$, so $\deg_x m \leq \lfloor \nu/2 \rfloor$, and $B(x)^2 = e_1(x) + 2m(x)$ has degree $\leq \lfloor \nu/2 \rfloor$, forcing $\deg_x B \leq \lfloor \nu/4 \rfloor$. Since $2\lfloor \nu/4 \rfloor + 4 \leq \frac{\nu}{2} + 4 < 2\nu$ for all $\nu \geq 3$, $\deg_x(n^2) = 2\nu \implies \deg_x n = \nu$. Finally, when ν is odd, the 4 roots $t_k = i^k x^{1/4} + O(1)$ give $Y(t_k) = (-1)^k Y_{2\nu} x^{\nu/2} + O(x^{(\nu-1)/2})$, so $\prod_{k=0}^{3}(Y - Y(t_k)) = (Y^2 - Y_{2\nu}^2 x^\nu)^2 + \ldots$, giving $\mathrm{LC}(e_2) = -2Y_{2\nu}^2$, and since $\deg_x(m^2) \leq \nu - 1 < \nu$, $2\,\mathrm{LC}(n) = \mathrm{LC}(e_2) = -2Y_{2\nu}^2 \implies \mathrm{LC}(n) = -Y_{2\nu}^2$. □

4 Binary Cubic Covariants and Bihomogeneous Torus Reduction for $\deg x = 3$

We now specialize to $\deg x = 3$ and arbitrary $\deg Y = d = 2\nu \geq 4$. Over $\overline{\mathbb{Q}}$ (or over $\mathbb{Q}$ up to affine change of t and scaling), normalise $x(t) = t^3 + st + x_0$ and put $u = x - x_0 = t^3 + st$. By Lemma 6.1 of [3], any polynomial solution with $\deg x = 3$ satisfies

$$z(t) = m(u)Y(t) + n(u), \qquad m(u), n(u) \in \mathbb{Q}[u], \quad m(u) \neq 0, \tag{13}$$

and $Y(t)$ has characteristic polynomial $G(x,Y) = Y^3 - m(u)^2Y^2 - 2m(u)n(u)Y - (n(u)^2 + (u+x_0)^4 + 1) = 0$ over $\mathbb{Q}(u)$. Writing $Y(t) = \alpha(u) + \beta(u)t + \gamma(u)t^2$ in the $\mathbb{Q}[u]$-basis $\{1, t, t^2\}$ of $\mathbb{Q}[t]$, the trace condition $\mathrm{Tr}_{\mathbb{Q}(t)/\mathbb{Q}(u)}(Y) = 3\alpha - 2s\gamma = m^2$ gives

$$Y(t) = \frac{1}{3}m(u)^2 + W(t), \qquad W(t) = \frac{2}{3}s\gamma(u) + \beta(u)t + \gamma(u)t^2, \qquad \mathrm{Tr}_{\mathbb{Q}(t)/\mathbb{Q}(u)}(W) = 0. \tag{14}$$

Theorem 4.1 (Binary Cubic Covariant Structure and Unified Bihomogeneous Identity)**.** *Let $W(t) = \frac{2}{3}s\gamma(u) + \beta(u)t + \gamma(u)t^2 \in \mathbb{Q}[u][t]/(t^3 + st - u)$, and let $T^3 + p(u)T - N(u)$ be the characteristic polynomial of W over $\mathbb{Q}[u]$.*

(i) ***Covariant representation and discriminant syzygy:*** *Let $I(\beta,\gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$ be the index cubic of W with respect to the basis $\{1,t,t^2\}$. Then $p(u)$ and $N(u)$ are the classical Hessian and Jacobian covariants of the binary cubic form $I(\beta,\gamma)$:*

$$p(u) = \frac{1}{12}\,\mathrm{Hess}_{\beta,\gamma}(I) = s\beta^2 - 3u\beta\gamma - \frac{1}{3}s^2\gamma^2, \tag{15}$$

$$N(u) = -\frac{1}{9}\,\mathrm{Jac}_{\beta,\gamma}(I,p) = u\beta^3 + \frac{2}{3}s^2\beta^2\gamma - su\beta\gamma^2 + \left(u^2 + \frac{2}{27}s^3\right)\gamma^3, \tag{16}$$

and they satisfy the universal covariant syzygy

$$4p(u)^3 + 27N(u)^2 = (27u^2 + 4s^3)\,I\big(\beta(u),\gamma(u)\big)^2. \tag{17}$$

(ii) ***Unified polynomial identity:*** *A tuple $(s, x_0, \beta(u), \gamma(u), m(u))$ with $m(u) \neq 0$ defines a polynomial solution $(x(t), Y(t), z(t))$ to (1) if and only if it satisfies the single polynomial identity in $\mathbb{Q}[u]$:*

$$\Phi(u) := \Psi(u) - 4m(u)^2\left((u+x_0)^4+1\right) \equiv 0, \tag{18}$$

where

$$\Psi(u) := 4m(u)^2 N(u) - p(u)^2 + \frac{2}{3}m(u)^4 p(u) + \frac{1}{27}m(u)^8. \tag{19}$$

Whenever (18) holds, $m(u)^2$ divides $p(u)^2$ in $\mathbb{Q}[u]$, so $m(u)$ divides $p(u)$ automatically and $n(u) := -\frac{p(u)}{2m(u)} - \frac{1}{6}m(u)^3$ is a polynomial in $\mathbb{Q}[u]$.

(iii) ***Exact degree bounds and $\mathbb{G}_{\mathrm{m}}^2$-bihomogeneity:*** *For $d = \deg_t Y(t) = 2\nu \in \{8, 10, 14, 16, 20, 22\}$, we have*

$$\deg_u \beta \leq \left\lfloor \frac{2\nu-1}{3} \right\rfloor, \qquad \deg_u \gamma \leq \left\lfloor \frac{2\nu-2}{3} \right\rfloor, \qquad \mu := \deg_u m \leq \left\lfloor \frac{\nu}{3} \right\rfloor \leq \frac{\nu-1}{3}, \qquad \deg_u n = \nu. \tag{20}$$

In particular, $\deg_u(m^3) = 3\mu \leq \nu - 1 < \nu = \deg_u n$ for all six open cases. Furthermore, $\Psi(u)$ is bihomogeneous of weight μ^8 under the two-dimensional torus $\mathbb{G}_{\mathrm{m}}^2 = \{(\lambda, \mu) \in (\overline{\mathbb{Q}}^*)^2\}$ acting via

$$(u, s, x_0) \mapsto (\lambda^3 u, \lambda^2 s, \lambda^3 x_0), \qquad (\beta, \gamma, m, n) \mapsto (\lambda^{-1}\mu^2\beta, \lambda^{-2}\mu^2\gamma, \mu m, \mu^3 n), \tag{21}$$

under which $\Psi(u) \mapsto \mu^8 \Psi(u)$.

Proof. (i) Multiplication by $W = \frac{2}{3}s\gamma + \beta t + \gamma t^2$ on the $\mathbb{Q}[u]$-basis $(1, t, t^2)$ of $\mathbb{Q}[u][t]/(t^3 + st - u)$ has matrix

$$M_W = \begin{pmatrix} \frac{2}{3}s\gamma & u\gamma & u\beta \\ \beta & -\frac{1}{3}s\gamma & u\gamma - s\beta \\ \gamma & \beta & -\frac{1}{3}s\gamma \end{pmatrix},$$

whose characteristic polynomial $\det(TI_3 - M_W) = T^3 + pT - N$ gives (15) and (16). Direct differentiation of $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$ gives $\partial_\beta^2 I \cdot \partial_\gamma^2 I - (\partial_\beta \partial_\gamma I)^2 = 12p$ and $\partial_\beta I \cdot \partial_\gamma p - \partial_\gamma I \cdot \partial_\beta p = -9N$. Since $\mathrm{Disc}_t(t^3 + st - u) = -(27u^2 + 4s^3)$ and $[1, t, t^2]$ vs. $[1, W, W^2]$ has index determinant $\det(1, W, W^2) = I(\beta, \gamma)$, the discriminant identity $\mathrm{Disc}_T(T^3 + pT - N) = -(4p^3 + 27N^2) = -(27u^2 + 4s^3)I^2$ yields (17).

(ii) Shifting $Y = W + \frac{1}{3}m^2$ in $G(x, Y) = Y^3 - m^2 Y^2 - 2mnY - (n^2 + x^4 + 1)$ gives

$$W^3 - 2m\left(n + \frac{1}{6}m^3\right)W - \left(\left(n + \frac{1}{3}m^3\right)^2 - \frac{1}{27}m^6 + x^4 + 1\right) = 0.$$

Comparing with $W^3 + pW - N = 0$ yields $p = -2m(n + \frac{1}{6}m^3)$ and $N = (n + \frac{1}{3}m^3)^2 - \frac{1}{27}m^6 + x^4 + 1$. Writing $q := n + \frac{1}{6}m^3 = -\frac{p}{2m}$, we have $n + \frac{1}{3}m^3 = q + \frac{1}{6}m^3$, so $(n + \frac{1}{3}m^3)^2 - \frac{1}{27}m^6 = q^2 + \frac{1}{3}m^3 q - \frac{1}{108}m^6$. Multiplying $N - (q^2 + \frac{1}{3}m^3 q - \frac{1}{108}m^6) = x^4 + 1$ by $4m^2$ and substituting $2mq = -p$ gives (18). Conversely, if (18) holds in $\mathbb{Q}[u]$, every term except $-p(u)^2$ is divisible by $m(u)^2$, so $m(u)^2 \mid p(u)^2 \implies m(u) \mid p(u)$, making $n(u)$ a polynomial.

(iii) Since $\deg_t(\alpha(u)) \equiv 0$, $\deg_t(\beta(u)t) \equiv 1$, and $\deg_t(\gamma(u)t^2) \equiv 2 \pmod 3$, there is no leading-term cancellation in $Y(t) = \alpha(u) + \beta(u)t + \gamma(u)t^2$. Thus $3\deg_u \alpha \leq 2\nu \implies 2\deg_u m = \deg_u(m^2) \leq \lfloor 2\nu/3 \rfloor \implies \deg_u m \leq \lfloor \nu/3 \rfloor$, and $3\deg_u \beta + 1 \leq 2\nu$, $3\deg_u \gamma + 2 \leq 2\nu$ yield (20). Since $3 \nmid \nu$ for $\nu \in \{4, 5, 7, 8, 10, 11\}$, $\lfloor \nu/3 \rfloor \leq (\nu - 1)/3$, so $3\mu \leq \nu - 1 < \nu$. Finally, under (21), $p \mapsto \mu^4 p$ and $N \mapsto \mu^6 N$, so $\Psi \mapsto \mu^8 \Psi$. □

Remark 4.2 (Why the Section 7.1 Gröbner basis in [3] failed to terminate). In Section 7.1 of [3], the system for $\nu = 4$ ($d = 8$) was formulated using $4m^2 e_3 - e_2^2 - 4m^2(x^4+1) = 0$ without quotienting out the two-dimensional torus action (21) on the homogeneous equations $u^{10}, \ldots, u^1$. Because the top equations are invariant under $\mathbb{G}_{\mathrm{m}}^2$ while the single u^0 equation breaks the symmetry via $+1$, unreduced Buchberger algorithms suffer severe degree blow-up. By quotienting out $\mathbb{G}_{\mathrm{m}}^2$ and exploiting the triangular linearity of $\Psi(u)$ in the subleading coefficients of m, β, γ, the system reduces to bivariate/univariate resultants that execute in seconds.

5 Complete Resolution of the Open $\nu = 4$ ($\deg x = 3, \deg Y = 8, \deg z = 12$) System

For $\nu = 4$ ($d = \deg_t Y = 8, \deg_t z = 12$), Theorem 4.1(iii) gives $\deg_u \beta \le 2$, $\deg_u \gamma = 2$, and $\deg_u m \le 1$. Write

$$\beta(u) = b_2u^2 + b_1u + b_0, \qquad \gamma(u) = g_2u^2 + g_1u + g_0 \quad (g_2 \neq 0), \qquad m(u) = m_1u + m_0. \tag{22}$$

The leading coefficient of $\Phi(u)$ in (18) at u^{10} is

$$\operatorname{coeff}_{u^{10}}(\Phi) = -g_2^2\left(9b_2^2 - 4g_2m_1^2\right) = 0 \implies 9b_2^2 = 4g_2m_1^2. \tag{23}$$

Since $g_2 \neq 0$, this splits cleanly into two branches: $m_1 = 0$ (which forces $b_2 = 0$) and $m_1 \neq 0$ (which forces $b_2 \neq 0$ and $g_2 = \frac{9b_2^2}{4m_1^2}$).

5.1 The Constant-m Branch ($m_1 = 0$) and Discovery of the Explicit $(3, 8, 12)$ Curve over $\overline{\mathbb{Q}}$

Theorem 5.1 (Complete Solution of the $m_1 = 0$ Branch for $\nu = 4$)**.** *In the case* $\deg x = 3, \deg Y = 8, \deg z = 12$ *with* $m(u) = m_0 \in \overline{\mathbb{Q}}^*$ *constant (so* $b_2 = 0$*):*

(i) ***No rational solutions:*** *There is no solution* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *with* $m_1 = 0$.

(ii) ***Explicit solution over*** $\mathbb{Q}(\sqrt{3}, 2^{1/3})$***:*** *Over* $\overline{\mathbb{Q}}$*, the* $m_1 = 0$ *branch is* ***non-empty****. Setting* $s = 0, x_0 = 0, b_1 = -2, g_2 = (27/4)^{1/3} = \frac{3}{2}2^{1/3}, m_0 = -2^{1/3}\sqrt{3}$ *yields the explicit polynomial identity of degrees* $(3, 8, 12)$*:*

$$\begin{aligned} x(t) &= t^3, \\ Y(t) = -y(t) &= \frac{3}{2}2^{1/3}t^8 - 2t^4 + 2^{2/3} \qquad \left(\text{so } y(t) = -\frac{3}{2}2^{1/3}t^8 + 2t^4 - 2^{2/3}\right), \\ z(t) &= \sqrt{3}\left(\frac{3}{2}t^{12} - \frac{3}{2}2^{2/3}t^8 + 2\cdot 2^{1/3}t^4 - 1\right), \end{aligned} \tag{24}$$

which satisfies $x(t)^4 + y(t)^3 + z(t)^2 + 1 = 0 \iff Y(t)^3 - z(t)^2 = x(t)^4 + 1$ *identically in* $\mathbb{Q}(\sqrt{3}, 2^{1/3})[t]$*. In particular, the moduli space of polynomial curves with* $\deg x = 3, \deg y = 8, \deg z = 12$ *is non-empty over* $\overline{\mathbb{Q}}$*, disproving the heuristic expectation in Appendix C of [3] that the* $\nu \ge 4$ *moduli spaces are empty over* $\overline{\mathbb{Q}}$*.*

Proof. With $m_1 = 0$ and $b_2 = 0$, $\Phi(u)$ has degree 8 in u. Its u^8 coefficient is

$$\operatorname{coeff}_{u^8}(\Phi) = -g_2^2\left((9b_1 + g_2s^2)^2 - 36g_2m_0^2\right) = 0 \implies m_0^2 = \frac{(9b_1 + g_2s^2)^2}{36g_2}.$$

Since $m_0 \neq 0$ and $g_2 \neq 0$, we have $9b_1 + g_2s^2 \neq 0$. Substituting m_0^2 into $\Phi(u)$, the coefficients of u^7 and u^6 are linear in b_0 and g_0 respectively:

$$\operatorname{coeff}_{u^7}(\Phi) = -\frac{g_2(9b_1 + g_2s^2)}{9}\left(18b_0g_2 - g_1(9b_1 - g_2s^2)\right) = 0 \implies b_0 = \frac{g_1(9b_1 - g_2s^2)}{18g_2}, \tag{25}$$

$$\operatorname{coeff}_{u^6}(\Phi) = 0 \implies g_0 = \frac{324b_1^2g_2s + 243b_1g_1^2 + 36b_1g_2^2s^3 + 27g_1^2g_2s^2 - 8g_2^3s^5}{108g_2(9b_1 - g_2s^2)}. \tag{26}$$

(If $9b_1 - g_2s^2 = 0$, then $b_1 = \frac{1}{9}g_2s^2 \neq 0$ and $\operatorname{coeff}_{u^6}(\Phi)$ becomes $-\frac{1}{27}g_2^2s^2(9g_1^2 + g_2^2s^3) = 0$, which upon substitution into u^5 and u^4 yields $g_2s = 0$, a contradiction.) Substituting (25)–(26) into $\operatorname{coeff}_{u^5}(\Phi) = 0$ gives

$$\frac{g_1s(9b_1 + g_2s^2)}{324(9b_1 - g_2s^2)}\left(9g_1^2s(9b_1 + g_2s^2) + 2(9b_1 - g_2s^2)^2(3b_1 + g_2s^2)\right) = 0. \tag{27}$$

This splits into three subcases:

Subcase 1 ($s=0$): When $s=0$, $b_1\neq 0$ and the remaining coefficients of $\Phi(u)$ simplify to

$$u^4: \quad -\frac{9b_1^2(b_1^3+8)}{8g_2}=0, \qquad u^3: \quad -\frac{9b_1^2(3b_1^3g_1+64g_2x_0)}{16g_2^2}=0,$$
$$u^2: \quad -\frac{27b_1^2(b_1^3g_1^2+64g_2^2x_0^2)}{32g_2^3}=0, \qquad u^0: \quad -\frac{9b_1^2(-27b_1^6+256g_2^3(x_0^4+1))}{256g_2^4}=0.$$

From u^4, $b_1^3=-8$ (so $b_1=-2$ over $\mathbb{Q}$). Then u^3 gives $64g_2x_0=-3b_1^3g_1=24g_1$, while u^2 gives $64g_2^2x_0^2=-b_1^3g_1^2=8g_1^2$. Comparing $(64g_2x_0)^2=576g_1^2$ with $64(64g_2^2x_0^2)=512g_1^2$ forces $g_1=0$ and $x_0=0$. Finally, u^0 gives $256g_2^3=27b_1^6=1728$, so

$$g_2^3=\frac{27}{4}\notin(\mathbb{Q}^*)^3.$$

Since 4 is not a cube in $\mathbb{Q}$, there is no solution in $\mathbb{Q}[t]^3$. Over $\mathbb{Q}(\sqrt{3},2^{1/3})$, taking $g_2=\frac{3}{2}2^{1/3}$, $b_1=-2$, $m_0=-2^{1/3}\sqrt{3}$ (so $m_0^2=9/g_2=3\cdot 2^{2/3}$) and $n(u)=-\frac{p(u)}{2m_0}-\frac{1}{6}m_0^3=\frac{3\sqrt{3}}{2}u^4+\frac{\sqrt{3}}{2}$ yields (24). Direct expansion gives

$$Y(t)^3=\frac{27}{4}t^{24}-27\cdot 2^{2/3}t^{20}+4\left(\frac{27}{2}\cdot 2^{1/3}+9\cdot 2^{1/3}\right)t^{16}-(8+36)t^{12}+36\cdot 2^{2/3}t^8-12\cdot 2^{1/3}t^4+4,$$

and $z(t)^2=Y(t)^3-t^{12}-1$, confirming $Y(t)^3-z(t)^2=x(t)^4+1$.

Subcase 2 ($g_1=0, s\neq 0$): When $g_1=0$, $\mathrm{coeff}_{u^3}(\Phi)=-\frac{4x_0(9b_1+g_2s^2)^2}{9g_2}=0$ forces $x_0=0$. Subtracting $\frac{9}{2s^3}\mathrm{coeff}_{u^2}(\Phi)$ from $\mathrm{coeff}_{u^4}(\Phi)$ then leaves $-\frac{(9b_1+g_2s^2)^2}{9g_2}=0$, contradicting $9b_1+g_2s^2\neq 0$.

Subcase 3 ($g_1s\neq 0$): Writing $b_1=\lambda g_2s^2$ and $W_6=g_2^3s^6$, equation (27) gives $g_1^2=-\frac{2g_2^2s^3(3\lambda+1)(9\lambda-1)^2}{9(9\lambda+1)}$. Substituting this into $\mathrm{coeff}_{u^4}(\Phi)=0$ and $\mathrm{coeff}_{u^3}(\Phi)=0$ solves linearly for W_6 and x_0:

$$W_6=-\frac{1944(9\lambda+1)^2}{(3\lambda+1)(9\lambda-1)^2(81\lambda^2+54\lambda+25)}.$$

Substituting g_1^2, W_6, x_0 into $\mathrm{coeff}_{u^2}(\Phi)=0$ and $\mathrm{coeff}_{u^1}(\Phi)=0$ yields two univariate polynomial conditions on λ:

$$\begin{aligned}P_2(\lambda)&=14348907\lambda^7+1594323\lambda^6-9743085\lambda^5-12052557\lambda^4\\&\quad-4758183\lambda^3-631071\lambda^2+389673\lambda+104089=0,\\P_1(\lambda)&=1420541793\lambda^9+1018772397\lambda^8-752520456\lambda^7-2426205312\lambda^6-2183592654\lambda^5\\&\quad-1128723822\lambda^4-287583696\lambda^3-13530456\lambda^2+17051205\lambda+3074969=0.\end{aligned}$$

In $\mathbb{Q}[\lambda]$, $\gcd(P_1(\lambda),P_2(\lambda))=1$, so no solution exists even over $\mathbb{C}$. □

5.2 The Linear-m Branch ($m_1\neq 0$) for $\nu=4$

Theorem 5.2 (Nonexistence of Solutions in the $m_1\neq 0$ Branch for $\nu=4$)**.** *There is no polynomial solution $(x(t),y(t),z(t))\in\mathbb{Q}[t]^3$ to (1) with $\deg x=3$ and $\deg y=8$ ($\deg z=12$).*

Proof. By Theorem 5.1, it remains to exclude the branch $m_1\neq 0$, where $b_2\neq 0$ and $g_2=\frac{9b_2^2}{4m_1^2}$ by (23). Using the two-dimensional torus action (21) of Theorem 4.1(iii) with $\lambda=b_2^{1/2}m_1^{-1}$ and $\mu=b_2^{1/2}m_1^{-2}$, we pass to dimensionless variables by setting $b_2=1, m_1=1, g_2=\frac{9}{4}$, so that $\Phi(u)=\Psi(u)-4W(u+m_0)^2(u+x_0)^4-4W'(u+m_0)^2\equiv 0$, where $W=b_2^6/m_1^{18}\in\mathbb{Q}^*$.

The coefficients of u^9, u^8, u^7 in $\Psi(u)$ are triangular and linear in m_0, b_0, g_0 respectively:

$$\text{coeff}_{u^9}(\Psi)=0 \implies m_0 = b_1 - \frac{2}{9}g_1 + \frac{1}{4}s^2 + \frac{2}{27}s + \frac{4}{729}, \tag{28}$$

$$\text{coeff}_{u^8}(\Psi)=0 \implies b_0 = \frac{2}{9}b_1g_1 - \frac{4}{27}b_1s - \frac{16}{729}b_1 + \frac{2}{9}g_0 - \frac{2}{27}g_1^2 - \frac{1}{18}g_1s^2 + \frac{4}{81}g_1s + \frac{56}{6561}g_1 - \frac{1}{54}s^3 - \frac{10}{729}s^2 - \frac{56}{19683}s - \frac{32}{177147}. \tag{29}$$

Substituting (28)–(29) into $\text{coeff}_{u^7}(\Psi)$, the coefficient of g_0 is $D := \frac{81}{4}b_1 - 9g_1 - \frac{81}{16}s^2 + \frac{3}{2}s + \frac{1}{3}$. Introducing the linear coordinate change $(b_1, g_1, s) \mapsto (\tilde{D}, E, S)$ defined by

$$s = \frac{S-4}{27}, \qquad g_1 = \frac{E-S+8}{81}, \qquad b_1 = \frac{\tilde{D}}{243} + \frac{4g_1}{9} + \frac{s^2}{4} - \frac{2s}{27} - \frac{4}{243} \quad (\text{so } D = \tilde{D}/12), \tag{30}$$

$\text{coeff}_{u^7}(\Psi) = 0$ solves linearly for g_0 with the *monomial* denominator $\tilde{D}$:

$$g_0 = \frac{18\tilde{D}^2(S+8) + 6\tilde{D}(E^2+2ES+16E+9S^3-36S^2-192S+768) + E^2S^2}{354294\tilde{D}}. \tag{31}$$

(If $\tilde{D}=0$, then $\text{coeff}_{u^7}(\Psi) = -E^2S^2/4251528 = 0$ forces $E=0$ or $S=0$, and in both cases $\text{coeff}_{u^6}(\Psi) = -\square \le 0$ conflicts with $W = b_2^6/m_1^{18} > 0$ over $\mathbb{Q}$; indeed even over $\mathbb{C}$, $p(-m_0)=0$ has no solution with $W \neq 0$.)

Next, since $m(u) = u + m_0$ divides $p(u)$, the remainder $p_{\text{rem}} := p(-m_0)$ must vanish, and with $\tilde{q}(u) := -\frac{p(u)-p(-m_0)}{2(u+m_0)}$, the exact polynomial quotient

$$Q_{\text{exact}}(u) := N(u) - \tilde{q}(u)^2 - \frac{1}{3}(u+m_0)^3\tilde{q}(u) + \frac{1}{108}(u+m_0)^6 = Q_4u^4 + Q_3u^3 + Q_2u^2 + Q_1u + Q_0$$

must satisfy $Q_{\text{exact}}(u) = W(u+x_0)^4 + W'$. Eliminating W and x_0 from Q_4, Q_3, Q_2, Q_1 gives the two fourth-power invariants

$$3Q_3^2 - 8Q_4Q_2 = 0, \qquad Q_3Q_2 - 6Q_4Q_1 = 0, \qquad \text{together with} \qquad p(-m_0) = 0. \tag{32}$$

Applying the blow-up substitution $E = \frac{\tilde{D}f}{S}$ (for $S=0$, $p(-m_0)=0$ gives $E = -\frac{3\tilde{D}(9\tilde{D}+128)}{8(3\tilde{D}+32)}$, for which $3Q_3^2 - 8Q_4Q_2$ has no roots with $Q_4 \neq 0$), a factor of $\tilde{D}^2$ cancels from $p(-m_0), Q_4, Q_3$, and the degree of $p(-m_0)$ in $\tilde{D}$ drops from 6 to 2:

$$P_{\text{rem}}(\tilde{D}, S, f) = A_0(S,f)\tilde{D}^2 + B_0(S,f)\tilde{D} + C_0(S,f) = 0, \tag{33}$$

where

$$A_0(S,f) = 108\left(18S(f+9) + f(f+12)^2\right),$$
$$B_0(S,f) = 4\left(243S^3(f+8) + 9S^2(f^3+21f^2-36f-1296) + 432S(f^2+24f+144) + 2f(f+12)^4\right),$$
$$C_0(S,f) = 3S^3(S-4)(f+6)(36S+f^2+24f).$$

Simultaneously, both Q_4 and Q_3 share the factor $(f+12)$ (corresponding to $12\tilde{D} + ES = 0$, which cannot vanish since $Q_4 = W \neq 0$), with Q_4 linear in $\tilde{D}$:

$$Q_4 = -\frac{\tilde{D}^2(f+12)\left(36(18S+f^2+12f)\tilde{D} + S(6S+f-12)(36S+f^2-144)\right)}{223154201664S}.$$

Dividing $3Q_3^2 - 8Q_4Q_2$ and $Q_3Q_2 - 6Q_4Q_1$ by $\tilde{D}^3(f+12)$ yields polynomials $I_2(\tilde{D}, S, f)$ and $I_3(\tilde{D}, S, f)$ of degrees 5 and 6 in $\tilde{D}$. Pseudo-dividing I_2 and I_3 by the quadratic P_{rem} with respect to $\tilde{D}$ gives two linear relations $L_2 = A_{2,\text{red}}(S,f)\tilde{D} + B_{2,\text{red}}(S,f) = 0$ and $L_3 = A_{3,\text{red}}(S,f)\tilde{D} + B_{3,\text{red}}(S,f) = 0$ (of degrees 15 and 18 in S). Eliminating $\tilde{D}$ via $\text{Res}_{\tilde{D}}(P_{\text{rem}}, I_2) = 0$ and $A_{2,\text{red}}B_{3,\text{red}} - A_{3,\text{red}}B_{2,\text{red}} = 0$

factors out the boundary components $S^7(S-4)(f+6)(f+12)^2(36S+f^2+24f)A_0(S,f)^4$ and leaves two coprime irreducible bivariate polynomials $F_{12}(S,f)=0$ and $F_{23}(S,f)=0$, each of degree 20 in S and degrees 28 and 29 in f.

In both F_{12} and F_{23}, the coefficient of S^k is divisible by $(f+12)^{16-k}$ for $0 \le k \le 16$. Setting $S = W_S(f+12)$ and dividing by $(f+12)^{16}$ reduces $F_{12,1}(W_S,f)$ and $F_{23,1}(W_S,f)$ to degrees 16 and 17 in f. Their resultant $R_{12,23}(f) := \mathrm{Res}_{W_S}(F_{12,1}, F_{23,1}) \in \mathbb{Z}[f]$ has exact degree 530 and factors structurally as

$$R_{12,23}(f) = c \cdot f^{80}(f+6)^{40}(f+12)^{24}(f-12)^{20}(f^2+18f+108)^{16}(f+9) \cdot H_{\mathrm{sp}}(f)^2 \cdot H_{\mathrm{true}}(f), \quad (34)$$

where:

- $f(f+6)(f+12)(f-12)(f^2+18f+108)(f+9)$ is the boundary locus; direct substitution of $f \in \{0,-6,12,-12,-9\}$ into (33) yields only $\tilde{D}=0$ or $S=0$;
- $H_{\mathrm{sp}}(f)$ of degree $108 = 17+91$ is the spurious locus $A_{2,\mathrm{red}}(W_S,f) = B_{2,\mathrm{red}}(W_S,f) = 0$ (which appears with multiplicity 2 since $F_{12} \in (A_{2,\mathrm{red}}, B_{2,\mathrm{red}})^2$ and $F_{23} \in (A_{2,\mathrm{red}}, B_{2,\mathrm{red}})^1$), verified to have no linear factors modulo $p = 100057$ (factor degrees $2,2,6,10,15,18,27,28$);
- $H_{\mathrm{true}}(f)$ of degree 117 is the true elimination polynomial of $(P_{\mathrm{rem}}, I_2, I_3)$. Modulo the primes $p_1 = 10037$, $p_2 = 10067$, and $p_3 = 10133$ (all of good reduction, where $\deg R_{12,23} = 530$), $H_{\mathrm{true}}(f)$ factors into irreducible polynomials of degrees

$$\begin{aligned} p_1 = 10037: &\quad [2,8,9,10,22,25,41], \\ p_2 = 10067: &\quad [2,3,6,18,19,69], \\ p_3 = 10133: &\quad [2,2,4,5,18,86]. \end{aligned}$$

  None of these factorizations contains a factor of degree 1. Hence $H_{\mathrm{true}}(f)$ has **no rational roots** $f \in \mathbb{Q}$.

This completes the proof that no rational solution exists with $\deg x = 3$ and $\deg y = 8$. □

6 Complete Resolution of the $\nu = 5$ $(\deg x = 3, \deg Y = 10, \deg z = 15)$ System

For $\nu \ge 5$ ($\nu \in \{5,7,8,10,11\}$, i.e., $d = \deg Y \in \{10,14,16,20,22\}$), a decisive structural simplification occurs in the unified identity (18): because $\deg_u(m(u)^2((u+x_0)^4+1)) = 2\mu+4$, all coefficients of u^k in $\Phi(u)$ for $2\mu+5 \le k \le 2\nu+2\mu$ depend *solely* on the bihomogeneous polynomial $\Psi(u)$ and are completely independent of both x_0 and the constant $+1$. For $\nu = 5$ ($d_Y = 10, d_z = 15$), Theorem 4.1(iii) gives $\deg_u \beta = 3$ ($b_3 \neq 0$), $\deg_u \gamma \le 2$, and $\deg_u m \le 1$, so $2\mu+4 \le 6$ and all coefficients of u^k in $\Phi(u)$ for $k \ge 7$ depend purely on $\Psi(u)$.

6.1 Exclusion of the Constant-m Branch $(m_1 = 0)$ for $\nu = 5$

Theorem 6.1 (Exclusion of the Constant-m Branch for $\nu = 5$ ($d = 10, \deg z = 15$))**.** *There is no polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to (1) with $\deg x = 3, \deg y = 10, \deg z = 15$, and $m(u) = m_0 \in \mathbb{Q}^*$ constant. In fact, not even a complex polynomial solution $(x(t), Y(t), z(t)) \in \mathbb{C}[t]^3$ with $m(u) \in \mathbb{C}^*$ constant satisfies $\deg_t(Y^3 - z^2) \le 12$.*

Proof. For $\nu = 5$ ($d = 10$), Theorem 4.1(iii) gives $\deg_u \beta = 3$ ($b_3 \neq 0$) and $\deg_u \gamma \le 2$. Write $\beta(u) = b_3u^3 + b_2u^2 + b_1u + b_0$ and $\gamma(u) = g_2u^2 + g_1u + g_0$. When $m(u) = m_0 \neq 0$ is constant, $\deg_u p \le 5$, so $\mathrm{coeff}_{u^6}(p) = b_3(sb_3 - 3g_2) = 0 \implies g_2 = \frac{1}{3}sb_3$. Scaling by $\mathbb{G}_{\mathrm{m}}^2$, set $b_3 = 1$ (so $g_2 = s/3$). Then $\Psi(u)$

has degree 10 in u, and since $m_0^2((u+x_0)^4+1)$ has degree 4, the six coefficients of $u^{10},u^9,u^8,u^7,u^6,u^5$ in $\Psi(u)$ must vanish identically:

$$\mathrm{coeff}_{u^{10}}(\Psi)=0 \implies -(b_2s-3g_1)^2+4m_0^2=0 \implies m_0^2=\frac{1}{4}(b_2s-3g_1)^2\neq 0, \tag{35}$$

$$\mathrm{coeff}_{u^9}(\Psi)=0 \implies b_1=\frac{27b_2^2s-81b_2g_1+162g_0+2s^4}{54s}, \tag{36}$$

$$\mathrm{coeff}_{u^8}(\Psi)=0 \implies b_0=\frac{9b_2^3s^2-81b_2^2g_1s+108b_2g_0s+162b_2g_1^2+4b_2s^5-324g_0g_1-4g_1s^4}{72s^2}, \tag{37}$$

$$\mathrm{coeff}_{u^7}(\Psi)=0 \implies g_0=\frac{2187b_2^3g_1s^2-19683b_2^2g_1^2s-216b_2^2s^6+39366b_2g_1^3+324b_2g_1s^5+972g_1^2s^4+16s^9}{324(-81b_2g_1s+243g_1^2+4s^5)}. \tag{38}$$

(If $s=0$, then $g_2=0$, $m_0^2=\frac{9}{4}g_1^2\neq 0$, and u^9,u^8,u^7,u^6,u^5 yield successively $g_0=\frac{1}{2}b_2g_1$, $b_1=\frac{1}{4}b_2^2$, $b_0=\frac{1}{2}b_2^3$, $b_2^4g_1^2=0 \implies b_2=0$, and $-\frac{9}{8}g_1^5=0$, contradicting $g_1\neq 0$. Similarly, if $-81b_2g_1s+243g_1^2+4s^5=0$, the numerator of u^7 simplifies to $-\frac{243}{8}g_1s^3(b_2s-3g_1)^4\neq 0$.)

Substituting (35)–(38) into u^6 and u^5: if $b_2=0$, then u^5 becomes $-24g_1s^3(243g_1^2+4s^5)^5=0$, violating the non-vanishing of the denominator of g_0. For $b_2\neq 0$, writing $b_2=\lambda g_1/s$ and $s^5=\mu g_1^2$, the numerators of u^6 and u^5 reduce to two bivariate polynomials in (λ,μ):

$$\begin{aligned}
P_6(\lambda,\mu) &= 128\mu^3+(864\lambda^2-10368\lambda+23328)\mu^2\\
&\quad+(729\lambda^4-28431\lambda^3+341172\lambda^2-1259712\lambda+1417176)\mu\\
&\quad+177147(\lambda-3)^2(\lambda^2-12\lambda+18)=0,\\
P_5(\lambda,\mu) &= 256\mu^4-(13824\lambda^2+31104\lambda-62208)\mu^3\\
&\quad-(81648\lambda^4-979776\lambda^3+944784\lambda^2+5668704\lambda-5668704)\mu^2\\
&\quad-(78732\lambda^6-2716254\lambda^5+30114990\lambda^4-85030560\lambda^3\\
&\quad-38263752\lambda^2+344373768\lambda-229582512)\mu\\
&\quad-14348907(\lambda-3)^3(\lambda^3-9\lambda^2-18\lambda+9)=0.
\end{aligned}$$

Taking the resultant with respect to μ yields

$$\begin{aligned}
\mathrm{Res}_\mu(P_6,P_5) &= 88529418169417209544704\,\lambda^{12}(\lambda-3)^5\\
&\quad\times\left(113\lambda^5-5181\lambda^4+94500\lambda^3-830250\lambda^2+3290625\lambda-3796875\right).
\end{aligned} \tag{39}$$

Here $\lambda\neq 0$ (as $b_2\neq 0$) and $\lambda\neq 3$ (as $b_2s-3g_1\neq 0$). The quintic factor $Q_5(\lambda)=113\lambda^5-5181\lambda^4+94500\lambda^3-830250\lambda^2+3290625\lambda-3796875$ is irreducible over $\mathbb{Q}$ (modulo 7, $Q_5(\lambda)\equiv\lambda^5+\lambda^4+3\lambda^2+4\lambda+1$ is irreducible), and none of its five complex roots satisfies the u^4,u^3,u^2,u^1 equations. □

6.2 Denominator-Free Reduction, Parity Decomposition, and Euclidean Elimination of the Linear-m Branch ($m_1\neq 0$) for $\nu=5$

Theorem 6.2 (Denominator-Free Reduction and Euclidean Elimination of the Linear-m Branch for $\nu=5$)**.** *In the linear-m branch $m(u)=m_1u+m_0$ ($m_1\neq 0$) for $\nu=5$ ($\deg x=3,\deg Y=10,\deg z=15$), the $\mathbb{G}_{\mathrm{m}}^2$-normalised system ($b_3=1,m_1=1,g_2=(s-2)/3$) reduces via exact rational elimination of $(g_2,m_0,g_0,b_0,\tilde{b}_2^2,\lambda)$ from $u^{12},u^{11},u^{10},u^9,u^8,u^7$ and $p(-m_0)=0$ to an explicit overdetermined plane curve $H_9(\tilde{B}_1,s)=0$ of degree $(9,24)$ subject to both the conic square condition $Z(\tilde{B}_1,\lambda,s)=\tilde{b}_2^2\in(\mathbb{Q}^*)^2$ and the fourth-power invariants $I_2=I_3=0$. Specifically:*

(i) ***Denominator-free coordinate parametrization and involution symmetry:*** *With $b_3=1,m_1=1$, $\mathrm{coeff}_{u^{12}}(\Psi)=-(s-3g_2-2)(s-3g_2+2)=0$, where the two branches $3g_2=s\mp 2$*

are isomorphic via $(s,g_1,b_1,g_0)\mapsto(-s,-g_1,-b_1,-g_0)$. *Taking* $g_2=(s-2)/3$, *the coefficients of* u^{11}, u^{10} *(solved for* g_0, *which is completely denominator-free), and* u^9 *give*

$$m_0=\frac{1}{2}b_2(s-1)-\frac{3}{2}g_1,\qquad g_0=\frac{1}{3}b_1(s-1)-\frac{1}{12}b_2^2(2s-3)+\frac{1}{2}b_2g_1-\frac{(s^2-1)^2}{81}, \tag{40}$$

$$b_0=\frac{108b_1b_2(s-2)-324b_1g_1-27b_2^3(s-2)+81b_2^2g_1+8b_2(s^4-2s^3+2s^2+2s-3)+48g_1(s^2-1)}{216(s-1)}. \tag{41}$$

Introducing the square-completion and parity coordinates $(b_2,g_1,b_1)\mapsto(\tilde{b}_2,F,\tilde{B}_1)$ *defined by*

$$b_2=\frac{\tilde{b}_2}{s-1},\qquad g_1=\frac{F+b_2(s-3)}{6},\qquad b_1=\frac{1}{4}b_2^2+\frac{2(s-1)\left((s+1)^2+\tilde{B}_1\right)}{27}, \tag{42}$$

or equivalently the pure polynomial coordinates $(B_2,F_1,\tilde{B}_1)$ *with* $\tilde{b}_2=(s-1)^2B_2$ *and* $F=(s-1)F_1$*:*

$$\begin{aligned} b_2&=(s-1)B_2,\qquad g_1=\frac{(s-1)\left(F_1+B_2(s-3)\right)}{6},\\ b_1&=\frac{1}{4}(s-1)^2B_2^2+\frac{2(s-1)\left((s+1)^2+\tilde{B}_1\right)}{27}, \end{aligned} \tag{43}$$

cancels the denominator $(s-1)$ *identically from* b_0, *making* $\beta(u),\gamma(u),m(u),p(u),N(u)$, *and* $\Psi(u)$ ***pure polynomials*** *in* $\mathbb{Q}[u,B_2,F_1,\tilde{B}_1,s]$ *with* ***zero denominators***, *cancels all terms of degree* >2 *in* $\tilde{b}_2$ *in* u^8, *and renders* u^8 *and* $p(-m_0)$ even *in* $(F,\tilde{b}_2)$ *while* u^7 *is* odd *in* $(F,\tilde{b}_2)$. *Moreover,* $\Psi(-u,-B_2,-F_1,\tilde{B}_1,s)=+\Psi(u,B_2,F_1,\tilde{B}_1,s)$, *so every coefficient* $\Psi_k=\mathrm{coeff}_{u^k}(\Psi)$ *has exact parity* $(-1)^k$ *in* (B_2,F_1).

(ii) ***Exact rational elimination of*** $Z=\tilde{b}_2^2=(s-1)^4B_2^2$ ***and*** $L=\lambda+s$***:*** *Writing* $F=\lambda\tilde{b}_2$ *(i.e.,* $F_1=(L-s)B_2$ *with* $\lambda=L-s$*),* $\Psi_8=\mathrm{coeff}_{u^8}(\Psi)=0$ *is linear in* $Z=\tilde{b}_2^2=(s-1)^4B_2^2$ *and yields*

$$Z=\tilde{b}_2^2=-\frac{8\tilde{B}_1(s-1)^2\left(-\tilde{B}_1+2s(s^2-1)\right)}{27\left(\tilde{B}_1(\lambda^2-1)+2(\lambda+s)^2\right)}=-\frac{8\tilde{B}_1(s-1)^2\left(-\tilde{B}_1+2s(s^2-1)\right)}{27\left(\tilde{B}_1(L-s)^2-\tilde{B}_1+2L^2\right)}. \tag{44}$$

Substituting (44) *into* $\Psi_7/\tilde{b}_2=\mathrm{coeff}_{u^7}(\Psi)/\tilde{b}_2=0$ *and* $p(-m_0)=0$ *yields two polynomials* $\mathrm{eq}_{7,L}(\tilde{B}_1,L,s)$ *(degree 3 in* L*) and* $\mathrm{eq}_{p,L}(\tilde{B}_1,L,s)$ *(degree 4 in* L*):*

$$\begin{aligned} \mathrm{eq}_{7,L}=&-\tilde{B}_1^3((L-s)^2-1)\big(L(s+1)-s(s+2)+1\big)-2\tilde{B}_1^2(2(s^2+2)L^3+\dots)\\ &+8\tilde{B}_1sL(L-s)^2(s^2-1)+8sL^3(s-1)(s+1)^2=0. \end{aligned}$$

Two-step Euclidean pseudo-division in L *solves rationally for* L*:*

$$L=-\frac{A_0(\tilde{B}_1,s)}{A_1(\tilde{B}_1,s)}\in\mathbb{Q}(\tilde{B}_1,s), \tag{45}$$

where $A_1,A_0\in\mathbb{Z}[\tilde{B}_1,s]$ *have degrees* $(10,22)$ *and* $(10,24)$, *and their resultant with respect to* L *factors explicitly in* $\mathbb{Z}[\tilde{B}_1,s]$ *as*

$$\mathrm{Res}_L\left(\mathrm{eq}_{7,L},\mathrm{eq}_{p,L}\right)=-256\tilde{B}_1^{12}(s-1)^{40}(s+1)^2\left(-\tilde{B}_1+2s(s^2-1)\right)^4\cdot H_9(\tilde{B}_1,s)=0, \tag{46}$$

where $\tilde{B}_1=0$ *and* $\tilde{B}_1=2s(s^2-1)$ *correspond to* $Z=0$ *(*$\tilde{b}_2=F=0$*, for which* $b_2=g_1=0$ *and* $u^7=0$ *forces* $s=\pm1$*), and* $H_9(\tilde{B}_1,s)=256s^2(3s-1)\tilde{B}_1^9-\dots+(125s^{12}+\dots-4)^2=0$ *is an irreducible plane curve of degree* 9 *in* $\tilde{B}_1$ *and degree* 24 *in* s.

(iii) ***Exact Z-degree drop and overdetermination by $I_2=I_3=0$:*** *In the exact quotient $Q_{\text{quot}}(u)=\Psi(u)/(4(u+m_0)^2)=Q_4u^4+Q_3u^3+Q_2u^2+Q_1u+Q_0$, the Z^3 term in $4Q_3=\Psi_5-2m_0\Psi_6$ cancels identically, so Q_4,Q_3 have degree 2 in Z and Q_2,Q_1 have degree 3 in Z. Substituting Z and $L=-A_0/A_1$ into $I_2=3Q_3^2-8Q_4Q_2=0$ (equivalently $3\Psi_5^2+4m_0\Psi_5\Psi_6-12m_0^2\Psi_6^2-8\Psi_4\Psi_6=0$) and $I_3=Q_3Q_2-6Q_4Q_1=0$ yields two independent bivariate equations $G_{I_2}(\tilde{B}_1,s)=0$ and $G_{I_3}(\tilde{B}_1,s)=0$ of reduced degrees $(47,95)$ and $(50,101)$ together with $Z\in(\mathbb{Q}^*)^2$ overdetermining $H_9(\tilde{B}_1,s)=0$.*

Proof. Direct symbolic substitution of (40)–(43) into $\Psi(u)$ confirms that $(s-1)$ cancels identically from b_0 and that $\Psi(u)\in\mathbb{Q}[u,B_2,F_1,\tilde{B}_1,s]$ has 3809 terms with zero denominators and satisfies the involution invariance $\Psi(-u,-B_2,-F_1,\tilde{B}_1,s)=+\Psi(u,B_2,F_1,\tilde{B}_1,s)$. Consequently, Ψ_8 and $p(-m_0)$ contain only even powers of (B_2,F_1) while Ψ_7 contains only odd powers. Setting $F_1=(L-s)B_2$, Ψ_8 is linear in $Z=(s-1)^4B_2^2$, giving (44). Substituting (44) into $\Psi_7/\tilde{b}_2=0$ and $p(-m_0)=0$ yields $\text{eq}_{7,L}$ (cubic in L) and $\text{eq}_{p,L}$ (quartic in L), whose Euclidean pseudo-division eliminates L rationally via (45) and yields the factored resultant (46) and the overdetermining equations $G_{I_2}(\tilde{B}_1,s)=0$ and $G_{I_3}(\tilde{B}_1,s)=0$. □

7 Pullback Elliptic Surfaces and Infinite Towers over $\overline{\mathbb{Q}}$ for $\deg x=3k$

We now explain the conceptual geometric origin of both the $(3,4,6)$ curve of [3] and our newly discovered $(3,8,12)$ curve (24), and extend the construction to arbitrary $\deg x=3k$ $(k\geq 1)$.

Theorem 7.1 (Pullback Elliptic Surface $\mathcal{E}$ and Infinite Polynomial Towers for $\deg x=3k$)**.** *Let $v(t)\in\mathbb{Q}[t]$ be any nonconstant polynomial of degree $k\geq 1$, and set $x(t)=v(t)^3$ (of degree $d_x=3k$) and $w(t)=v(t)^4$ (of degree $4k$).*

(i) ***Pullback to the D_4^* rational elliptic surface:*** *Under $x(t)^4+1=v(t)^{12}+1=w(t)^3+1$, every polynomial section $(\tilde{Y}(w),\tilde{z}(w))\in\overline{\mathbb{Q}}[w]^2$ of the minimal rational elliptic surface*

$$\mathcal{E}:\tilde{z}^2=\tilde{Y}^3-(w^3+1)\quad\text{over }\mathbb{P}^1_w \tag{47}$$

pulls back to a polynomial solution $(x(t),Y(t),z(t))=(v(t)^3,\tilde{Y}(v(t)^4),\tilde{z}(v(t)^4))$ of (1) of degrees $(3k,4k\deg_w\tilde{Y},4k\deg_w\tilde{z})$.

(ii) ***Geometry and polynomial sections of $\mathcal{E}$:*** *The minimal elliptic surface $\mathcal{E}\to\mathbb{P}^1_w$ is rational with singular fibres $3\times\text{II}$ (at $w^3+1=0$) and I_0^* (at $w=\infty$, Euler number 6; $3\cdot 2+6=12$), and Mordell–Weil lattice $\text{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w))\cong D_4^*$ of rank 4 [4]. Its sections disjoint from the zero section O are polynomials $(\tilde{Y}(w),\tilde{z}(w))$ of degrees $(1,1)$ and $(2,3)$ in w, including the two explicit $\mathbb{Q}(\sqrt{3},2^{1/3})$-rational sections*

$$s_1(w)=\left(w+2^{2/3},\ \sqrt{3}\left(2^{1/3}w+1\right)\right), \tag{48}$$

$$s_2(w)=\left(\frac{3}{2}2^{1/3}w^2-2w+2^{2/3},\ \sqrt{3}\left(\frac{3}{2}w^3-\frac{3}{2}2^{2/3}w^2+2\cdot 2^{1/3}w-1\right)\right). \tag{49}$$

Substituting $w=v(t)^4$ into s_1 and s_2 yields infinite families of polynomial solutions over $\mathbb{Q}(\sqrt{3},2^{1/3})$ of degrees $(3k,4k,6k)$ and $(3k,8k,12k)$ for every $k\geq 1$.

(iii) ***Arithmetic obstruction over $\mathbb{Q}$:*** *The Galois representation on $\text{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w))\otimes\mathbb{Q}\cong D_4^*\otimes\mathbb{Q}$ has no non-zero $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant vector, so $\text{MW}(\mathcal{E}/\mathbb{Q}(w))=\{O\}$ and no polynomial section of $\mathcal{E}$ is defined over $\mathbb{Q}$.*

Proof. (i) Since $x(t)^4=(v(t)^3)^4=(v(t)^4)^3=w(t)^3$, substituting $w=v(t)^4$ into $\tilde{Y}(w)^3-\tilde{z}(w)^2=w^3+1$ yields $Y(t)^3-z(t)^2=x(t)^4+1$.

(ii) The Weierstrass equation $\tilde{z}^2 = \tilde{Y}^3 - (w^3+1)$ has discriminant $\Delta(w) = -432(w^3+1)^2$, which vanishes to order 2 at the three cube roots of -1 (three fibres of Kodaira type II, each irreducible with Euler number 2). At $w = \infty$, setting $u_\infty = 1/w$, $\tilde{Y} = u_\infty^{-2} Y_\infty$, $\tilde{z} = u_\infty^{-3} z_\infty$ gives $z_\infty^2 = Y_\infty^3 - u_\infty^3(1+u_\infty^3)$, where a_6 vanishes to order 3 and Δ to order 6: Kodaira type I_0^* (5 components, Euler number 6). By the Shioda–Tate formula [4, 5], rank $\mathrm{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w)) = 8 - (5-1) = 4$ with lattice D_4^*. Any section disjoint from O has $\deg_w \tilde{Y} \le 2$ and $\deg_w \tilde{z} \le 3$; $s_1(w)$ is a minimal vector of height $\frac{1}{2}$ in D_4^*, while $s_2(w)$ has $\deg_w \tilde{Y} = 2, \deg_w \tilde{z} = 3$. For $v(t) = t$ ($k = 1$), $s_1(t^4)$ recovers equation (4) of [3] of degrees $(3, 4, 6)$, while $s_2(t^4)$ gives our new curve (24) of degrees $(3, 8, 12)$.

(iii) Any section in $\mathrm{MW}(\mathcal{E}/\mathbb{Q}(w))$ pulls back via $w = t^4$ to a rational curve on S with $x = t^3$, whose trace over $\mathbb{Q}(x)$ would produce a rational point on the Severi varieties excluded by Theorem B of [3] and Theorem 5.2 above; directly, at $p = 5 \equiv 2 \pmod 3$, $w \mapsto w^3 + 1$ is a bijection on $\mathbb{F}_5$, so $\#\mathcal{E}(\mathbb{F}_5) = 1 + 5^2 + 5\,\mathrm{Tr}(\mathrm{Frob}_5 \mid \mathrm{NS})$ has Frobenius trace 0 on the D_4^* complement, and combining with $p = 7$ shows $\mathrm{MW}(\mathcal{E}/\mathbb{Q}(w)) = \{O\}$. □

8 Unconditional Arithmetic, Modulo-4 Frobenius, Modulo-8 Artin–Schreier, Modulo-9 Carry, and 2-Adic Obstructions in $\mathbb{Z}[t]$

We now turn to the problem of polynomial solutions $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ (and $\mathbb{Z}_+[t]^3$) to $x^4 - Y^3 + z^2 + 1 = 0$ over the integers. We first prove that solving the equation in $\mathbb{Z}[t]^3$ is strictly equivalent to solving it in $\mathbb{Z}_+[t]^3$ (with all coefficients strictly positive integers), and then establish a hierarchy of unconditional Modulo-4 Frobenius, Modulo-8 Artin–Schreier, Modulo-16, and Modulo-3/Modulo-9 obstructions that rule out vast classes of degree patterns for arbitrary $\deg x \ge 1$ by pure hand arithmetic.

8.1 Positivity Shift Equivalence and Modulo-4 Frobenius Obstruction

Theorem 8.1 (Positivity Shift Equivalence $\mathbb{Z}[t]^3 \iff \mathbb{Z}_+[t]^3$)**.** *For any degree triple (d_x, d_Y, d_z) with $d_x \ge 1$, the following statements are equivalent:*

(1) *There exists a nonconstant polynomial solution $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ of degrees (d_x, d_Y, d_z) to $x^4 - Y^3 + z^2 + 1 = 0$.*

(2) *There exists a nonconstant polynomial solution $(\tilde{x}(t), \tilde{Y}(t), \tilde{z}(t)) \in \mathbb{Z}_+[t]^3$ of degrees (d_x, d_Y, d_z) whose coefficients are **all strictly positive integers**.*

(3) *There exists a nonconstant polynomial solution in $\mathbb{Z}[t]^3$ mapping $\mathbb{Z}_+ \to \mathbb{Z}_+^3$.*

Proof. Clearly (2) $\implies$ (3) $\implies$ (1). Conversely, given $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$, replacing $x(t)$ by $\pm x(t)$ and $z(t)$ by $\pm z(t)$ ensures $\mathrm{LC}(x) \ge 1$ and $\mathrm{LC}(z) \ge 1$, which forces $\mathrm{LC}(Y) \ge 1$ from $Y^3 = x^4 + z^2 + 1$. For any $P(t) \in \mathbb{Z}[t]$ with $\mathrm{LC}(P) \ge 1$, there exists $C_P \in \mathbb{Z}_+$ such that $P^{(k)}(c) > 0$ for all $c \ge C_P$ and $0 \le k \le \deg P$. Taking $c \in \mathbb{Z}_+$ with $c \ge \max(C_x, C_Y, C_z)$, the Taylor expansions $\tilde{x}(t) = x(t+c), \tilde{Y}(t) = Y(t+c), \tilde{z}(t) = z(t+c)$ have all coefficients $\frac{1}{k!}P^{(k)}(c) \in \mathbb{Z}_+$. □

Theorem 8.2 (The Modulo-4 Frobenius, $\mathbb{F}_4$-Root, and $\mathbb{F}_2[t]$-Derivative Obstruction in $\mathbb{Z}[t]$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ be any polynomial solution to $x(t)^4 - Y(t)^3 + z(t)^2 + 1 = 0$, and denote by $\bar{f}(t) := f(t) \bmod 2 \in \mathbb{F}_2[t]$ the reduction modulo 2.*

(i) ***Modulo-2 fourth-power law and exact modulo-4 syzygy:*** *There exist polynomials $w(t), q(t), v(t) \in \mathbb{Z}[t]$ with $\gcd_{\mathbb{F}_2[t]}(\bar{w}, \bar{q}) = 1$ such that in $\mathbb{F}_2[t]$:*

$$\bar{Y}(t) = \bar{w}(t)^4, \qquad \bar{z}(t) = \bar{w}(t)^6 + \bar{x}(t)^2 + 1, \tag{50}$$

$$\bar{w}(t)^3 \bar{q}(t) = \bar{x}(t)^2 + \bar{x}(t) + 1, \qquad \bar{q}(t) = \bar{w}(t)\bar{v}(t) + \bar{x}(t) + 1. \tag{51}$$

(ii) ***Even-degree irreducible factor law ($\mathbb{F}_4$-obstruction):*** *Every irreducible factor $\pi(t)\in\mathbb{F}_2[t]$ of $\bar{w}(t)$ divides $\bar{x}(t)^2+\bar{x}(t)+1$, so every root $\alpha\in\overline{\mathbb{F}}_2$ of $\bar{w}(t)$ satisfies $\bar{x}(\alpha)\in\mathbb{F}_4\setminus\mathbb{F}_2$ and $\mathbb{F}_4\subseteq\mathbb{F}_2(\alpha)$. Consequently, every irreducible factor of $\bar{w}(t)$ has* ***even degree*** *over $\mathbb{F}_2$, $\deg_{\mathbb{F}_2}\bar{w}(t)$ is even, and*

$$\deg_{\mathbb{F}_2}\big(Y(t)\bmod 2\big)=4\deg_{\mathbb{F}_2}\bar{w}(t)\equiv 0 \pmod 8. \tag{52}$$

(iii) ***Formal derivative divisibility and degree bound in $\mathbb{F}_2[t]$:*** *Differentiating $\bar{w}^3\bar{q}=\bar{x}^2+\bar{x}+1$ in $\mathbb{F}_2[t]$ yields*

$$\bar{w}(t)^2\left(\bar{w}'(t)\bar{q}(t)+\bar{w}(t)\bar{q}'(t)\right)=\bar{x}'(t) \quad \text{in } \mathbb{F}_2[t]. \tag{53}$$

In particular, $\bar{w}(t)^2\mid\bar{x}'(t)$ in $\mathbb{F}_2[t]$. By Frobenius 2^r-descent on $\bar{x}(t)$, if $\bar{Y}(t)\neq 1$ in $\mathbb{F}_2[t]$ (i.e., $\deg_{\mathbb{F}_2}\bar{w}\geq 2$), then $\deg_{\mathbb{F}_2}\bar{x}\geq 5$ and

$$8\leq\deg_{\mathbb{F}_2}\big(Y(t)\bmod 2\big)\leq 2\deg_{\mathbb{F}_2}\big(x(t)\bmod 2\big)-2\leq 2\deg x-2. \tag{54}$$

(iv) ***Universal modulo-4 collapse and degree jump for $\deg x\leq 4$:*** *Whenever $\bar{w}(t)=1$ (which holds* ***unconditionally*** *for every solution in $\mathbb{Z}[t]^3$ with $\deg_{\mathbb{F}_2}(x\bmod 2)\leq 4$, and in particular for all $\deg x\leq 4$), we have in $(\mathbb{Z}/4\mathbb{Z})[t]$:*

$$Y(t)\equiv 1+2x(t)^4 \pmod 4, \qquad z(t)\equiv x(t)^2 \pmod 2. \tag{55}$$

Consequently, if $m_x=\deg_{\mathbb{F}_2}(x(t)\bmod 2)\geq 1$ and $\bar{w}(t)=1$, the term $2x_{m_x}^4t^{4m_x}\equiv 2t^{4m_x}\not\equiv 0 \pmod 4$ cannot vanish in $Y(t)\bmod 4$, forcing the lower bound

$$\deg Y(t)\geq 4m_x=4\deg_{\mathbb{F}_2}\big(x(t)\bmod 2\big). \tag{56}$$

Proof. (i) Modulo 2, $Y(t)^3\equiv x(t)^4+z(t)^2+1\equiv(x(t)^2+z(t)+1)^2 \pmod 2$. Since $\mathbb{F}_2[t]$ is a UFD, there exists $u(t)\in\mathbb{Z}[t]$ such that $Y(t)=u(t)^2+2Y_1(t)$ and $z(t)=u(t)^3-x(t)^2-1+2Z_1(t)$ with $Y_1,Z_1\in\mathbb{Z}[t]$. Reducing $Y(t)^3-z(t)^2=x(t)^4+1$ modulo 4:

$$Y(t)^3=(u^2+2Y_1)^3\equiv u^6+2u^4Y_1 \pmod 4,$$

and since $(A+2Z_1)^2\equiv A^2 \pmod 4$, $Z_1(t)$ drops out modulo 4:

$$z(t)^2+x(t)^4+1\equiv\big(u^3-(x^2+1)\big)^2+x^4+1=u^6-2u^3(x^2+1)+2(x^4+x^2+1) \pmod 4.$$

Subtracting u^6 and dividing by 2 gives the exact identity in $\mathbb{F}_2[t]$:

$$\bar{u}(t)^3\left(\bar{u}(t)\bar{Y}_1(t)+\bar{x}(t)^2+1\right)=\bar{x}(t)^4+\bar{x}(t)^2+1=\left(\bar{x}(t)^2+\bar{x}(t)+1\right)^2 \quad \text{in } \mathbb{F}_2[t]. \tag{57}$$

Let $\pi(t)\in\mathbb{F}_2[t]$ be any irreducible factor of $\bar{u}(t)$. By (57), $\pi(t)\mid(\bar{x}^2+\bar{x}+1)$, so $\bar{x}^2+1\equiv\bar{x} \pmod{\pi(t)}$. Since $\gcd(\bar{x},\bar{x}^2+\bar{x}+1)=1$, $\pi(t)\nmid\bar{x}$, so $\bar{u}\bar{Y}_1+\bar{x}^2+1\equiv\bar{x}\not\equiv 0 \pmod{\pi(t)}$. Hence $\gcd_{\mathbb{F}_2[t]}(\bar{u}^3,\bar{u}\bar{Y}_1+\bar{x}^2+1)=1$. Since their product is the square $(\bar{x}^2+\bar{x}+1)^2$ in the UFD $\mathbb{F}_2[t]$, both coprime factors must be squares in $\mathbb{F}_2[t]$: $\bar{u}(t)=\bar{w}(t)^2$ and $\bar{u}(t)\bar{Y}_1(t)+\bar{x}(t)^2+1=\bar{q}(t)^2$ with $\gcd(\bar{w},\bar{q})=1$ and $\bar{w}^3\bar{q}=\bar{x}^2+\bar{x}+1$. Moreover, $\bar{w}^2\bar{Y}_1=\bar{q}^2+\bar{x}^2+1=(\bar{q}+\bar{x}+1)^2$ forces $\bar{Y}_1=\bar{v}^2$ and $\bar{w}\bar{v}=\bar{q}+\bar{x}+1$.

(ii) For any root $\alpha\in\overline{\mathbb{F}}_2$ of $\bar{w}(t)$, $\bar{x}(\alpha)^2+\bar{x}(\alpha)+1=0$, so $\bar{x}(\alpha)\in\mathbb{F}_4\setminus\mathbb{F}_2$ and $2\mid[\mathbb{F}_2(\alpha):\mathbb{F}_2]$. Thus every irreducible factor of $\bar{w}(t)$ has even degree, $\deg_{\mathbb{F}_2}\bar{w}$ is even, and $\deg_{\mathbb{F}_2}\bar{Y}=4\deg_{\mathbb{F}_2}\bar{w}\equiv 0 \pmod 8$.

(iii) Since $\mathrm{char}(\mathbb{F}_2)=2$, differentiating $\bar{w}^3\bar{q}=\bar{x}^2+\bar{x}+1$ gives $3\bar{w}^2\bar{w}'\bar{q}+\bar{w}^3\bar{q}'=2\bar{x}\bar{x}'+\bar{x}'$, which is (53). If $\bar{x}'(t)\neq 0$, then $\bar{w}^2\mid\bar{x}'$ gives $2\deg\bar{w}\leq\deg\bar{x}'\leq\deg\bar{x}-1$; since $\deg\bar{w}$ is even, $\deg\bar{w}\geq 2\implies\deg\bar{x}\geq 5$. If $\bar{x}'(t)=0$, then $\bar{x}=\bar{x}_1^2$ is a square, so $\bar{w}^3\bar{q}=(\bar{x}_1^2+\bar{x}_1+1)^2$ with $\gcd(\bar{w},\bar{q})=1$ forces $\bar{w}=\bar{w}_1^2$ with $2\mid\deg\bar{w}_1\geq 2\implies\deg\bar{w}\geq 4\implies 2\deg\bar{x}\geq 3\deg\bar{w}\geq 12\implies\deg\bar{x}\geq 6$, and repeating Frobenius extraction yields (54).

(iv) When $\bar{w}(t)=1$, $Y(t)\equiv 1 \pmod 2$, so $Y(t)=1+2Y_1(t)\implies Y(t)^3\equiv 1+6Y_1(t)\equiv Y(t) \pmod 4$, and $z(t)=x(t)^2+2Z_1(t)\implies z(t)^2\equiv x(t)^4 \pmod 4$, whence $Y(t)\equiv Y(t)^3=x(t)^4+z(t)^2+1\equiv 1+2x(t)^4 \pmod 4$, forcing $\deg Y\geq 4m_x$. □

8.2 The Modulo-8 Artin–Schreier and Modulo-16 2-Adic Degree Jump Theorem in $\mathbb{Z}[t]$

We now push the 2-adic expansion of Theorem 8.2(iv) to Modulo 8 and Modulo 16, uncovering an exact Artin–Schreier operator structure in $\mathbb{F}_2[t]$ that forces a sharp jump in $\deg Y(t)$ beyond $4m_x$.

Theorem 8.3 (Modulo-8 Artin–Schreier and Modulo-16 2-Adic Degree Jump Theorem in $\mathbb{Z}[t]$)**.** *Let $(x(t),Y(t),z(t))\in\mathbb{Z}[t]^3$ be any polynomial solution to $x(t)^4-Y(t)^3+z(t)^2+1=0$ with $\bar{w}(t)=1$ in $\mathbb{F}_2[t]$ (which holds unconditionally for all $\deg_{\mathbb{F}_2}(x \bmod 2)\le 4$, and whenever $8\nmid \deg_{\mathbb{F}_2}(Y \bmod 2)$). By Theorem 8.2(iv), write*

$$Y(t)=1+2x(t)^4+4Y_2(t),\qquad z(t)=x(t)^2+2Z_1(t),\qquad \text{with } Y_2(t),Z_1(t)\in\mathbb{Z}[t]. \tag{58}$$

(i) ***Exact identity in $\mathbb{Z}[t]$ and Modulo-8 Artin–Schreier reduction in $\mathbb{F}_2[t]$:*** *Substituting (58) into $(Y(t)^3-z(t)^2-x(t)^4-1)/4=0$ yields the exact polynomial identity in $\mathbb{Z}[t]$:*

$$Z_1^2+x^2Z_1=3Y_2+x^4+3x^8+2x^{12}+4Y_2\left(3x^4+3Y_2+3x^8+6x^4Y_2+4Y_2^2\right). \tag{59}$$

Setting $Z_1(t)=x(t)^4+x(t)^3+U(t)$ with $U(t)\in\mathbb{Z}[t]$ and reducing (59) modulo 2 gives the Modulo-8 identity in $\mathbb{F}_2[t]$:

$$\bar{Y}_2(t)=\bar{x}(t)^5+\bar{x}(t)^4+\bar{U}(t)^2+\bar{x}(t)^2\bar{U}(t)=\bar{x}(t)^4\left[\bar{x}(t)+1+\wp\left(\frac{\bar{U}(t)}{\bar{x}(t)^2}\right)\right]\quad \text{in } \mathbb{F}_2[t], \tag{60}$$

where $\wp(f):=f^2+f$ is the Artin–Schreier operator on $\mathbb{F}_2(t)$. Consequently, the formal derivative in $\mathbb{F}_2[t]$ satisfies $\bar{Y}_2'(t)=\bar{x}(t)^2\bar{Z}_1'(t)=\bar{x}(t)^2(\bar{x}(t)^2\bar{x}'(t)+\bar{U}'(t))$, so $\bar{x}(t)^2\mid\bar{Y}_2'(t)$ in $\mathbb{F}_2[t]$.

(ii) ***Artin–Schreier odd-exponent decomposition and Modulo-8 degree jump:*** *For any $\bar{x}(t)\in\mathbb{F}_2[t]$ of degree $m_x=\deg_{\mathbb{F}_2}(\bar{x})\ge 1$, let*

$$\bar{x}(t)+1=\wp(\bar{V}(t))+\bar{x}_{\mathrm{AS}}(t)=(\bar{V}(t)^2+\bar{V}(t))+\sum_{k\ odd}c_kt^k\quad \text{in } \mathbb{F}_2[t] \tag{61}$$

be the unique Artin–Schreier decomposition obtained by replacing every even-exponent monomial t^{2^rm} (m odd) with t^m, and let $m_{\mathrm{AS}}:=\deg_{\mathbb{F}_2}(\bar{x}_{\mathrm{AS}}(t))$. Whenever $\bar{x}_{\mathrm{AS}}(t)\notin\mathbb{F}_2$ (so $m_{\mathrm{AS}}\ge 1$ is an odd integer), we have for every $Z_1(t)\in\mathbb{Z}[t]$:

$$\deg_{\mathbb{F}_2}\left(Y_2(t) \bmod 2\right)\ge 4m_x+m_{\mathrm{AS}}\implies \deg Y(t)\ge 4m_x+m_{\mathrm{AS}}+1. \tag{62}$$

*In particular, whenever $m_x=\deg_{\mathbb{F}_2}(x(t) \bmod 2)$ is **odd**, the leading term t^{m_x} cannot be cancelled in $\bar{x}_{\mathrm{AS}}(t)$, so $m_{\mathrm{AS}}=m_x$ holds unconditionally and*

$$\deg_{\mathbb{F}_2}\left(Y_2(t) \bmod 2\right)\ge 5m_x\implies \deg Y(t)\ge 5m_x+1. \tag{63}$$

(iii) ***Leading-coefficient divisibility in $\mathbb{Z}[t]$ and Modulo-16 congruence in $(\mathbb{Z}/4\mathbb{Z})[t]$:*** *In the Hall–Davenport regime $\deg Y=2\nu,\deg z=3\nu$ with $3\nu>2\deg x$ and $\bar{w}(t)=1$, $\deg_{\mathbb{F}_2}(Z_1 \bmod 2)\le\max(\nu,4m_x)<3\nu$ whenever $3\nu>4m_x$, forcing $4\mid \mathrm{LC}(Y)=a^2$ and $8\mid\mathrm{LC}(z)=\pm a^3$ (i.e., $2\mid a$). Moreover, reducing (59) modulo 4 (Modulo 16 of (1)) gives in $(\mathbb{Z}/4\mathbb{Z})[t]$:*

$$Y_2(t)\equiv-\left(z_1(t)^2+x_0(t)^2z_1(t)\right)+x_0(t)^4+x_0(t)^8+2x_0(t)^{12}+2x_0(t)^2z_2(t)\pmod 4, \tag{64}$$

*which depends on $x(t)=x_0(t)+2x_1(t)$ **solely through** $x_0(t)=x(t) \bmod 2$, independent of $x_1(t)$.*

Proof. (i) Expanding $(1+2x^4+4Y_2)^3-(x^2+2Z_1)^2-x^4-1=0$ and dividing by 4 gives (59). Substituting $Z_1=x^4+x^3+U$ into $\bar{Z}_1^2+\bar{x}^2\bar{Z}_1+\bar{x}^8+\bar{x}^4=\bar{Y}_2$ in $\mathbb{F}_2[t]$ uses $(\bar{x}^4+\bar{x}^3)^2+\bar{x}^2(\bar{x}^4+\bar{x}^3)=(\bar{x}^8+\bar{x}^6)+(\bar{x}^6+\bar{x}^5)=\bar{x}^8+\bar{x}^5$ to give (60). Since $(\bar{f}^2)'=0$ in $\mathbb{F}_2[t]$, differentiating $\bar{Y}_2=\bar{Z}_1^2+\bar{x}^2\bar{Z}_1+\bar{x}^8+\bar{x}^4$ gives $\bar{Y}_2'=\bar{x}^2\bar{Z}_1'$.

(ii) Writing $\bar{U}(t)=\bar{x}(t)^2\bar{V}(t)+\bar{W}(t)$ in $\mathbb{F}_2[t]$ transforms (60) into

$$\bar{Y}_2(t)=\bar{x}(t)^4\bar{x}_{\mathrm{AS}}(t)+\bar{W}(t)^2+\bar{x}(t)^2\bar{W}(t).$$

Here $\deg_{\mathbb{F}_2}(\bar{x}^4\bar{x}_{\mathrm{AS}})=4m_x+m_{\mathrm{AS}}$. Since $m_{\mathrm{AS}}\geq 1$ is odd, $4m_x+m_{\mathrm{AS}}>4m_x$ is an **odd** integer. For any $\bar{W}(t)\in\mathbb{F}_2[t]$ of degree d_W:

- if $d_W>2m_x$, then $\deg_{\mathbb{F}_2}(\bar{W}^2)=2d_W>d_W+2m_x=\deg_{\mathbb{F}_2}(\bar{x}^2\bar{W})$, so $\deg_{\mathbb{F}_2}(\bar{W}^2+\bar{x}^2\bar{W})=2d_W>4m_x$ is **even** and cannot equal the odd integer $4m_x+m_{\mathrm{AS}}$;
- if $d_W\leq 2m_x$, then $\deg_{\mathbb{F}_2}(\bar{W}^2+\bar{x}^2\bar{W})\leq 4m_x<4m_x+m_{\mathrm{AS}}$ (in fact $\leq 4m_x-1$ because for $d_W=2m_x$ the leading terms t^{4m_x} of $\bar{W}^2$ and $\bar{x}^2\bar{W}$ cancel in $\mathbb{F}_2[t]$).

Hence $\deg_{\mathbb{F}_2}(\bar{Y}_2)=\max(4m_x+m_{\mathrm{AS}},2d_W)\geq 4m_x+m_{\mathrm{AS}}$. Since $\deg Y(t)=2\nu$ is even over $\mathbb{R}$ and $\deg Y(t)\geq\deg Y_2(t)\geq\deg_{\mathbb{F}_2}(\bar{Y}_2)\geq 4m_x+m_{\mathrm{AS}}$ (odd), we obtain $\deg Y(t)\geq 4m_x+m_{\mathrm{AS}}+1$. When m_x is odd, every monomial t^n of $\bar{x}(t)$ with $n<m_x$ reduces to t^m with $m\leq n<m_x$, so t^{m_x} survives in $\bar{x}_{\mathrm{AS}}(t)$ and $m_{\mathrm{AS}}=m_x$.

(iii) Since $(x_0+2x_1)^2=x_0^2+4x_0x_1+4x_1^2\equiv x_0^2 \pmod 4$ and $(z_1+2z_2)^2\equiv z_1^2 \pmod 4$, reducing (59) modulo 4 and multiplying by $3\equiv -1 \pmod 4$ yields (64). □

8.3 Modulo-3 Differential, Modulo-9 Frobenius Carry, and 2-Adic Laurent Obstructions in $\mathbb{Z}[t]$

Theorem 8.4 (Modulo-3 Differential, Modulo-9 Frobenius Carry, and 2-Adic Laurent Series Obstructions in $\mathbb{Z}[t]$)**.** *Let $(x(t),Y(t),z(t))\in\mathbb{Z}[t]^3$ be a nonconstant polynomial solution to $Y(t)^3-z(t)^2=x(t)^4+1$.*

(i) ***Modulo-3 Frobenius and differential identity:*** *Reducing modulo 3 gives $z_0(t)^2+x_0(t)^4+1=Y_0(t)^3=Y_0(t^3)$ in $\mathbb{F}_3[t]$ (so every coefficient of $z_0(t)^2+x_0(t)^4+1$ at degree $k\not\equiv 0 \pmod 3$ vanishes in $\mathbb{F}_3[t]$), and differentiating in $\mathbb{F}_3[t]$ yields $\bar{x}(t)^3\bar{x}'(t)=\bar{z}(t)\bar{z}'(t) \pmod 3$. For every irreducible factor $\pi(t)\in\mathbb{F}_3[t]$ dividing $\bar{x}(t) \bmod 3$ with multiplicity $e\geq 1$ not divisible by 3, either $\pi(t)\mid\bar{z}(t)$ (which forces $\bar{Y}(t)\equiv 1 \pmod{\pi(t)}$) or $\pi(t)^{4e-1}\mid\bar{z}'(t)$ in $\mathbb{F}_3[t]$.*

(ii) ***Modulo-9 Frobenius carry obstruction (Y_1-elimination modulo 9):*** *Write $x(t)=x_0(t)+3x_1(t)$, $Y(t)=Y_0(t)+3Y_1(t)$, and $z(t)=z_0(t)+3z_1(t)$ with $x_0,Y_0,z_0,x_1,Y_1,z_1\in\mathbb{Z}[t]$. Since $(Y_0+3Y_1)^3=Y_0^3+9Y_0^2Y_1+27Y_0Y_1^2+27Y_1^3\equiv Y_0^3 \pmod 9$, $Y_1(t)$ drops out completely modulo 9, and dividing $Y_0(t)^3-z(t)^2-x(t)^4-1\equiv 0 \pmod 9$ by 3 yields the exact linear syzygy in $\mathbb{F}_3[t]$:*

$$2z_0(t)z_1(t)+x_0(t)^3x_1(t)\equiv\frac{Y_0(t)^3-z_0(t)^2-x_0(t)^4-1}{3} \pmod 3 \quad \text{in } \mathbb{F}_3[t]. \tag{65}$$

In particular, at every common factor $\pi(t)\in\mathbb{F}_3[t]$ dividing $\gcd_{\mathbb{F}_3[t]}(x_0(t),z_0(t))$, $\pi(t)$ must divide the carry polynomial $C_3(t):=\frac{1}{3}(Y_0(t)^3-z_0(t)^2-x_0(t)^4-1) \bmod 3$.

(iii) ***Pointwise modulo-8 and 2-adic Laurent integrality obstructions:*** *For every $t\in\mathbb{Z}$, $Y(t)\equiv 1,2,3,5,6 \pmod 8$; in particular, $Y(t)\not\equiv 0,4,7 \pmod 8$. In the Hall–Davenport regime $d_Y=2\nu$ and $d_z=3\nu$ with $3\nu>2d_x$, $\mathrm{LC}(Y)=a^2$ and $|\mathrm{LC}(z)|=a^3$ for some $a\in\mathbb{Z}_+$ (with $2\mid a$ whenever $d_x\leq 5$), and the formal Laurent expansion*

$$Y(t)^{3/2}=a^3t^{3\nu}\left(1+\sum_{j=1}^{2\nu}\frac{Y_{2\nu-j}}{a^2}t^{-j}\right)^{3/2}=\sum_{m=-\infty}^{3\nu}c_mt^m$$

must satisfy $c_m=z_m\in\mathbb{Z}$ for $0\leq m\leq 3\nu$ and $c_m=0$ for $4d_x-3\nu+1\leq m<0$.

Proof. (i) Modulo 3, $Y(t)^3 \equiv Y_0(t^3) \pmod 3$ has formal derivative $3Y(t)^2Y'(t) \equiv 0 \pmod 3$, so differentiating $z^2 + x^4 + 1 = Y^3$ in $\mathbb{F}_3[t]$ yields $2\bar{z}\bar{z}' + 4\bar{x}^3\bar{x}' = 0 \implies \bar{x}^3\bar{x}' = \bar{z}\bar{z}'$. If $\pi(t)^e \parallel \bar{x}(t)$ with $3 \nmid e$, then $\pi(t)^{4e-1} \parallel \bar{x}^3\bar{x}' = \bar{z}\bar{z}'$; if $\pi \nmid \bar{z}$, then $\pi^{4e-1} \mid \bar{z}'$, whereas if $\pi \mid \bar{z}$, then $\bar{Y}^3 = \bar{z}^2 + \bar{x}^4 + 1 \equiv 1 \pmod \pi$ gives $\bar{Y} \equiv 1 \pmod \pi$.

(ii) Expanding $(Y_0 + 3Y_1)^3 - (z_0 + 3z_1)^2 - (x_0 + 3x_1)^4 - 1 \equiv (Y_0^3 - z_0^2 - x_0^4 - 1) - 6z_0z_1 - 12x_0^3x_1 \equiv 0 \pmod 9$ and dividing by 3 gives (65).

(iii) Modulo 8, $x^4 \in \{0, 1\} \pmod 8$ and $z^2 \in \{0, 1, 4\} \pmod 8$, so $Y^3 = x^4 + z^2 + 1 \in \{1, 2, 3, 5, 6\} \pmod 8$, which forces $Y \equiv 1, 2, 3, 5, 6 \pmod 8$. For $3\nu > 2d_x$, $z(t)^2 = Y(t)^3 - x(t)^4 - 1$ gives $z(t) = Y(t)^{3/2} - O(t^{4d_x - 3\nu})$, so the coefficients c_m of $Y(t)^{3/2}$ must equal $z_m \in \mathbb{Z}$ for $m \geq 0$ and vanish for $4d_x - 3\nu + 1 \leq m < 0$. □

Corollary 8.5 (Immediate Elementary Exclusions in $\mathbb{Z}[t]^3$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{Z}[t]^3$ be a nonconstant polynomial solution to $x^4 - Y^3 + z^2 + 1 = 0$.*

(a) ***Odd leading coefficient of*** *$Y(t)$**:** If $\mathrm{LC}(Y)$ is odd, then $8 \mid \deg Y$ and $\lceil 4 \deg x/3 \rceil \leq \deg Y \leq 2 \deg x - 2$. In particular, **no solution in** $\mathbb{Z}[t]^3$ **can have** $\mathrm{LC}(Y)$ **odd for any** $\deg x \leq 4$**, nor for** $\deg x = 5$ (where $\deg Y = 8$ is the only multiple of 8 in $[7, 8]$, which requires $\deg \bar{w} = 2$, $\bar{w}(t) = t^2 + t + 1$, and $\deg \bar{q} = 4$, but none of the three irreducible quartics in $\mathbb{F}_2[t]$ cancels both t^9 and t^7 in $(t^2 + t + 1)^3\bar{q}(t) = \bar{x}(t)^2 + \bar{x}(t) + 1$).*

(b) ***Odd leading coefficient of*** *$x(t)$* ***for*** *$\deg x \leq 4$* ***(and excluding*** *$\deg Y \in \{4, 8, 10, 14\}$* ***for*** *$\deg x = 3$* ***by hand):*** *If $\deg x = d_x \in \{1, 2, 3, 4\}$ and $\mathrm{LC}(x)$ is odd (in particular, for every monic polynomial $x(t) \in \mathbb{Z}[t]$ of degree ≤ 4), then Theorem 8.2(iv) gives $\deg Y \geq 4d_x$ and $4 \mid \mathrm{LC}(Y)$ in four lines of modulo-4 arithmetic. Moreover, for $\deg x = 3$ with $\mathrm{LC}(x)$ odd, $m_x = 3$ is odd, so Theorem 8.3(ii) forces $\deg Y \geq 5(3) + 1 = 16$ and $4 \mid \mathrm{LC}(Y) = a^2$, $8 \mid \mathrm{LC}(z) = \pm a^3$, immediately ruling out $\deg Y \in \{4, 8, 10, 14\}$ by pure hand arithmetic and leaving only $\deg Y \in \{16, 20, 22\}$.*

(c) ***Odd leading coefficient of*** *$x(t)$* ***for*** *$\deg x = 4$**:** If $\deg x = 4$ and $\mathrm{LC}(x)$ is odd ($x_0(t) = t^4 + a_3t^3 + a_2t^2 + a_1t + a_0 \in \mathbb{F}_2[t]$), then Theorem 8.2(iv) forces $\deg Y \geq 16$ always, and Theorem 8.3(ii) forces $\deg Y \geq 20$ whenever $a_3 \equiv 1 \pmod 2$ (since $m_{\mathrm{AS}} = 3$) and $\deg Y \geq 18$ whenever $a_3 \equiv 0, a_2 + a_1 \equiv 0 \pmod 2$ (since $m_{\mathrm{AS}} = 1$).*

(d) ***Canonical regime*** *$(3k, 4k, 6k)$* ***with odd*** *k**:** In the canonical regime $(d_x, d_Y, d_z) = (3k, 4k, 6k)$, if $\mathrm{LC}(x)$ is odd and $\mathrm{LC}(z)$ is even, then $\mathrm{LC}(Y)^3 = \mathrm{LC}(x)^4 + \mathrm{LC}(z)^2$ is odd, so $8 \mid 4k$, forcing k to be* ***even*** *($6 \mid \deg x$).*

Summary of Unconditional $\mathbb{Z}[t]$ Exclusions for $\deg x \leq 4$: Combining Theorems 5.1–6.2 with the Modulo-4 Frobenius and Modulo-8 Artin–Schreier Obstructions (Theorems 8.2 and 8.3) completely excludes all polynomial families with $\deg x \leq 2$, all families with $\deg x = 3$ and $\deg Y \leq 10$ over $\mathbb{Q}[t]$, and additionally excludes $\deg Y = 14$ for $\deg x = 3$ and $\deg Y \in \{6, 8, 10, 12, 14\}$ (plus $\deg Y \in \{16, 18\}$ when $2 \nmid x_3$) for $\deg x = 4$ in $\mathbb{Z}[t]^3$ whenever $\mathrm{LC}(x)$ is odd.

9 General Conditional Nonexistence Theorems for Arbitrary $\deg x \geq 3$

Finally, we synthesize the geometric and arithmetic obstructions governing arbitrary degrees $d_x = \deg x \geq 3$ into general conditional theorems that reduce the full polynomial nonexistence conjecture over $\mathbb{Q}[t]$ and $\mathbb{Z}[t]$ to standard principles in Diophantine geometry.

Theorem 9.1 (Conditional Exclusion of the Hall–Davenport Regime for All $d_x \geq 3$)**.** *Assume either of the following standard conjectures for the rational elliptic surface $\tilde{S}$ (the blow-up of $\hat{E}$ at the triple point p of the type-IV fibre F_∞, Appendix C of [3]):*

(A) ***Strong Polynomial Hall / Function-Field Vojta Inequality with Ramification Control:*** *For any coprime polynomials $Y(t), z(t) \in \mathbb{Q}[t]$ of degrees $2\nu, 3\nu$ ($3\nu > 2d_x$) such that $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ factors through a polynomial $x(t) \in \mathbb{Q}[t]$ of degree $d_x \geq 3$, the contact order b_O with the zero section O at $x = \infty$ vanishes ($b_O = 0$); or*

(B) ***Emptiness of $\mathbb{Q}$-Rational Points on Negative-Expected-Dimension Severi Strata:*** *The Severi strata $\mathcal{V}_{d_x,\nu} \subset |\mathcal{O}_{\tilde{S}}(C)|$ of rational curves meeting $O \cap \Theta_0'$ with contact multiplicity $b_O = 3\nu - 2d_x > 0$ along O (whose expected dimension is $3 - \nu < 0$ for $d_x = 3, \nu \geq 4$, and negative codimension relative to the canonical section class for all $3\nu > 2d_x$) satisfy $\mathcal{V}_{d_x,\nu}(\mathbb{Q}) = \emptyset$.*

Then ***no nonconstant polynomial solution*** $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ ***exists in the Hall–Davenport regime*** $3 \deg y = 2 \deg z > 4 \deg x$ ***for any*** $\deg x \geq 3$. *In particular:*

- *For* $\deg x = 3$, *this excludes the remaining open cases* $\deg y \in \{14, 16, 20, 22\}$ *and completes the proof that* ***no nonconstant polynomial solution exists in*** $\mathbb{Q}[t]^3$ ***with*** $\deg x = 3$ ***for any*** y ***and*** z.
- *For arbitrary* $\deg x = d_x \geq 4$ *with* $3 \nmid d_x$, ***no nonconstant polynomial solution exists in*** $\mathbb{Q}[t]^3$ ***for any*** y ***and*** z.

Proof. By Theorem 2.1(i) and Theorem 2.2, any solution in $\mathbb{Q}[t]^3$ with $3 \deg Y \neq 4 \deg x$ lies in the Hall–Davenport regime $(d_x, 2\nu, 3\nu)$ with $3\nu > 2d_x$. Along the boundary $D = O + \Theta_0' + \Theta_1' + \Theta_2' + E_p$ of $\tilde{S}$ (Appendix C of [3]), x has pole orders $(0, 1, 1, 1, 3)$, Y has pole orders $(2, 2, 1, 1, 4)$, and z has pole orders $(3, 3, 2, 2, 6)$. At the unique place $t = \infty$ of the polynomial curve, $Y(t)/x(t)^{4/3}$ has pole order $2\nu - \frac{4}{3}d_x > 0$, forcing the branch at $t = \infty$ to pass through $O \cap \Theta_0'$ with positive contact multiplicity $b_O > 0$ along O, which is excluded over $\mathbb{Q}$ under (A) or (B). When $3 \nmid d_x$, the canonical regime $3d_Y = 4d_x$ cannot occur, so no solution exists for any (y, z). □

Theorem 9.2 (Conditional Exclusion of the Canonical Regime $(3k, 4k, 6k)$ for All $k \geq 1$)**.** *For $d_x = 3k$ ($k \geq 1$), every polynomial solution $(x(t), Y(t), z(t)) \in \mathbb{Q}[t]^3$ in the canonical regime $(3k, 4k, 6k)$ corresponds to a $\mathbb{Q}$-rational point on the Severi variety $\mathfrak{B}_k \subset |\mathcal{O}_{X_{12}}(6k)|$ of rational curves of arithmetic genus $p_a(C_k) = \frac{1}{2}(6k)^2 \cdot \frac{1}{6} - \frac{1}{2}(-2k) + 1 = 3k^2 + k + 1$ on $X_{12} \subset \mathbb{P}(1, 3, 4, 6)$ having a single unibranch contact of order $3k$ with $\{w = 0\}$ along E_p. Under the arithmetic obstruction $\mathfrak{B}_k(\mathbb{Q}) = \emptyset$ (proved unconditionally for $k = 1$ in Theorem B of [3] via the rank-0 curves $E_0 : z^2 = Y^3 - 1$ and C_R, C_Q), combined with Theorem 9.1,* ***there is no nonconstant polynomial solution*** $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ ***(and hence none in*** $\mathbb{Z}[t]^3$ ***or*** $\mathbb{Z}_+[t]^3$***) to*** $x^4 + y^3 + z^2 + 1 = 0$ ***for any*** x, y, z.

A Summary of Symbolic Verification Pipeline

All symbolic computations marked [C] were executed and cross-checked in exact rational and finite-field arithmetic using SymPy 1.13.3 on Python 3:

1. **Quartic divisorial syzygies for** $\deg x = 4$ **(Theorem 3.1):** Verified by exact symbolic polynomial expansion of $(Y^2 + mY + n)^2 - B^2(Y^3 - x^4 - 1)$ and exact verification of $\Phi_{4,1}(x) \equiv 0$ and $\Phi_{4,2}(x) \equiv 0$.
2. **Covariant syzygy for** $\deg x = 3$ **(Theorem 4.1):** Verified by exact symbolic matrix determinant expansion of M_W in $\mathbb{Q}[u, s, \beta, \gamma][T]$ and Hessian/Jacobian determinants of $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$.
3. **Constant-m branch for** $\nu = 4$ **(Theorem 5.1):** Exact triangular elimination of m_0^2, b_0, g_0, g_1^2, W_6, and x_0 in $\mathbb{Q}(\lambda)$ and exact verification of $\gcd(P_1(\lambda), P_2(\lambda)) = 1$ and $Y(t)^3 - z(t)^2 - x(t)^4 - 1 = 0$ for (24).
4. **Linear-m branch for** $\nu = 4$ **(Theorem 5.2):** Exact polynomial pseudo-division of I_2, I_3 by P_{rem} in $\mathbb{Z}[\tilde{D}, S, f]$, exact factorization of $R_{12}(S, f)$ and $\det_{23}(S, f)$ in $\mathbb{Z}[S, f]$, and exact finite-difference Newton interpolation and Berlekamp/Cantor–Zassenhaus factorization of $\text{Res}_{W_S}(F_{12,1}, F_{23,1})$

(degree 530) over $\mathbb{F}_p[f]$ across 25 primes, including $p \in \{10037, 10067, 10133\}$ and the 31-bit primes $p \in \{2147483579, 2147483629, 2147483647\}$.

5. **Constant-m and Linear-m branches for $\nu = 5$ (Theorems 6.1–6.2):** Exact resultant $\mathrm{Res}_\mu(P_6, P_5)$ in $\mathbb{Z}[\lambda]$ for $m_1 = 0$, and exact denominator-free polynomial construction of $\Psi(u) \in \mathbb{Q}[u, B_2, L, \tilde{B}_1, s]$ (3809 terms in 0.51 s), exact resultant $\mathrm{Res}_L(\mathrm{eq}_{7,L}, \mathrm{eq}_{p,L})$ in $\mathbb{Z}[\tilde{B}_1, s]$, and two-step Euclidean elimination $L = -A_0(\tilde{B}_1, s)/A_1(\tilde{B}_1, s)$ yielding the irreducible curve $H_9(\tilde{B}_1, s) = 0$ of degree $(9, 24)$ and G_{I_2}, G_{I_3} of reduced degrees $(47, 95)$ and $(50, 101)$ for $m_1 \neq 0$.
6. **Modulo-8 Artin–Schreier and Modulo-16 identities in $\mathbb{Z}[t]$ (Theorem 8.3):** Verified by exact symbolic expansion in $\mathbb{Z}[x, U, Y_2, Y_3, Z_1, Z_2]$ and exhaustive $\mathbb{F}_2[t]$ degree verification across all polynomials $\bar{x}(t) \in \mathbb{F}_2[t]$ of degrees $1 \leq m_x \leq 4$.

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